

Cluster Analysis – Beyond Basics

Peerapon S.

Machine Learning (1/67)

Topics

Hard Clustering ⇒ אבונטוורטונגן

Gaussian mixture model

percent
ה' אבונטוורטונגן
אין אונזערע פראגראםן
אין soft clustering

Hierarchical clustering

Density-based clustering

Gaussian Distribution

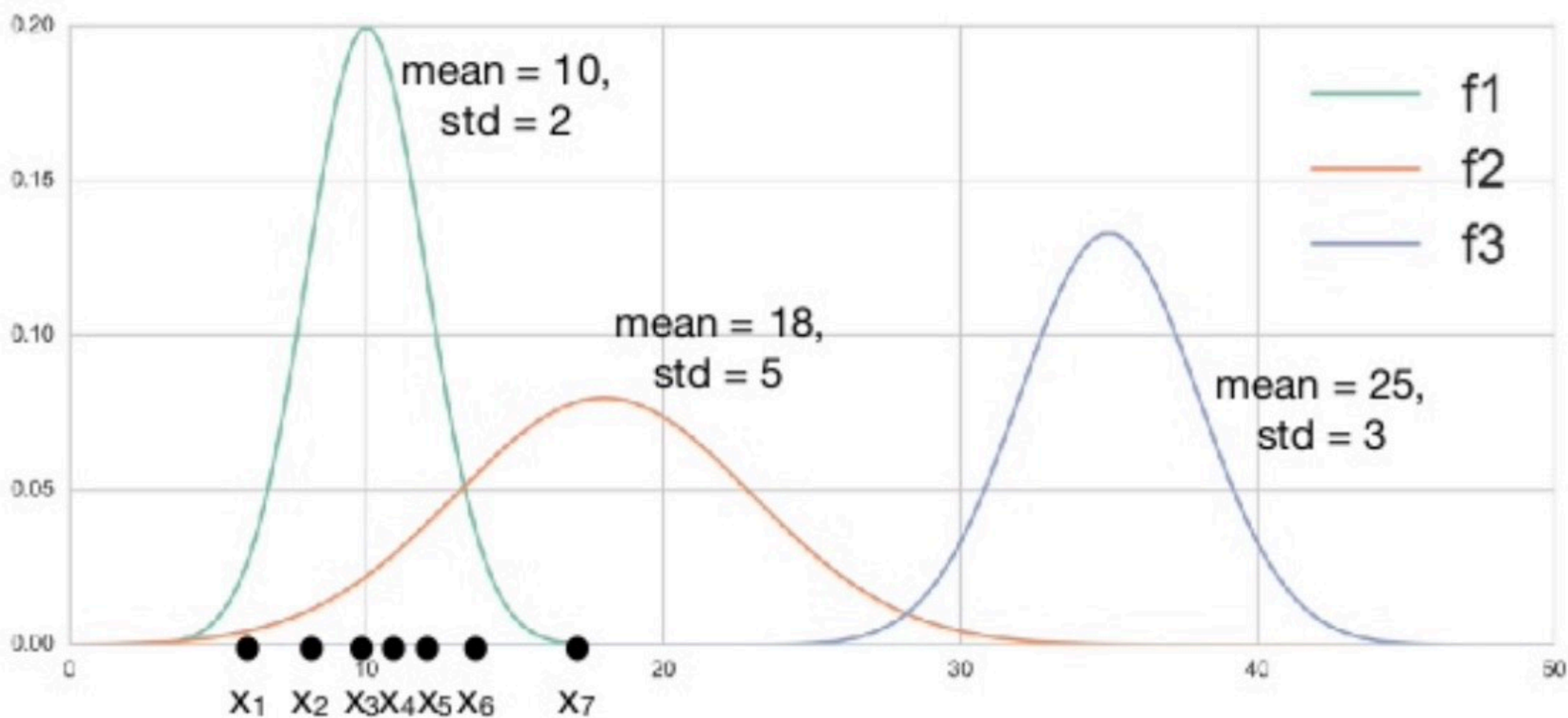
တွဲဝေမှုကိရိယာများကိုအသုံးပြု

Consider a univariate Gaussian distribution with parameters $\theta = (\mu, \sigma)$

x given θ cluster

$$f(x|\theta) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

S.D. *Variance*



Maximum Likelihood Estimation (MLE)

Given a set of 1D data $\mathbb{X} = \{\overbrace{x_1, x_2, \dots, x_n}^{\text{data point}}\}$, define the *likelihood function* and *log-likelihood function* as

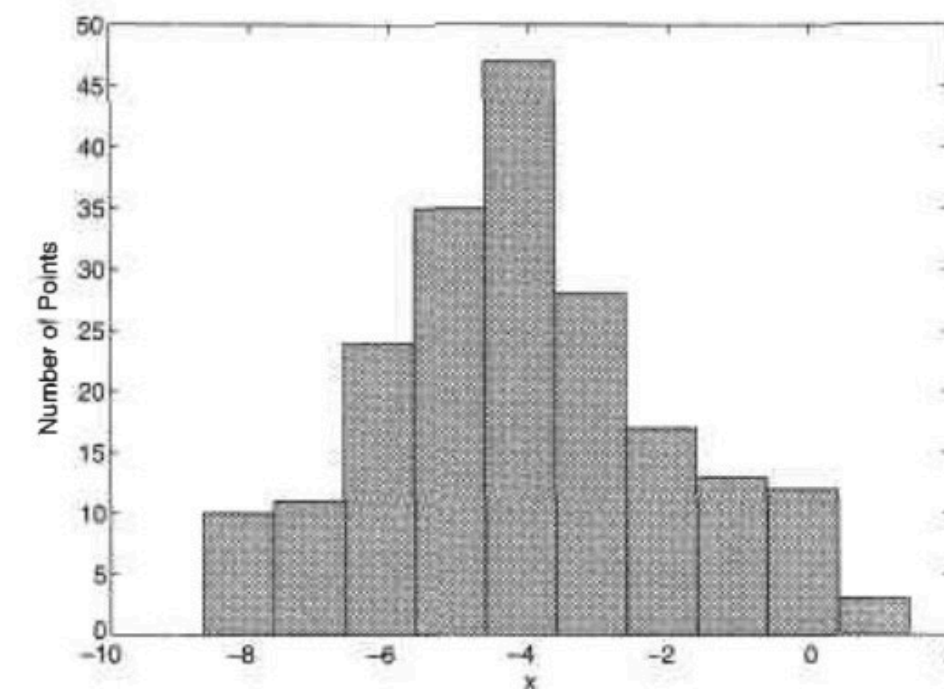
$$L(\theta|\mathbb{X}) = \prod_{i=1}^n f(x_i, \theta), \quad \theta = (\mu, \sigma)$$

$$= \prod_{i=1}^n \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x_i - \mu)^2}{2\sigma^2}}$$

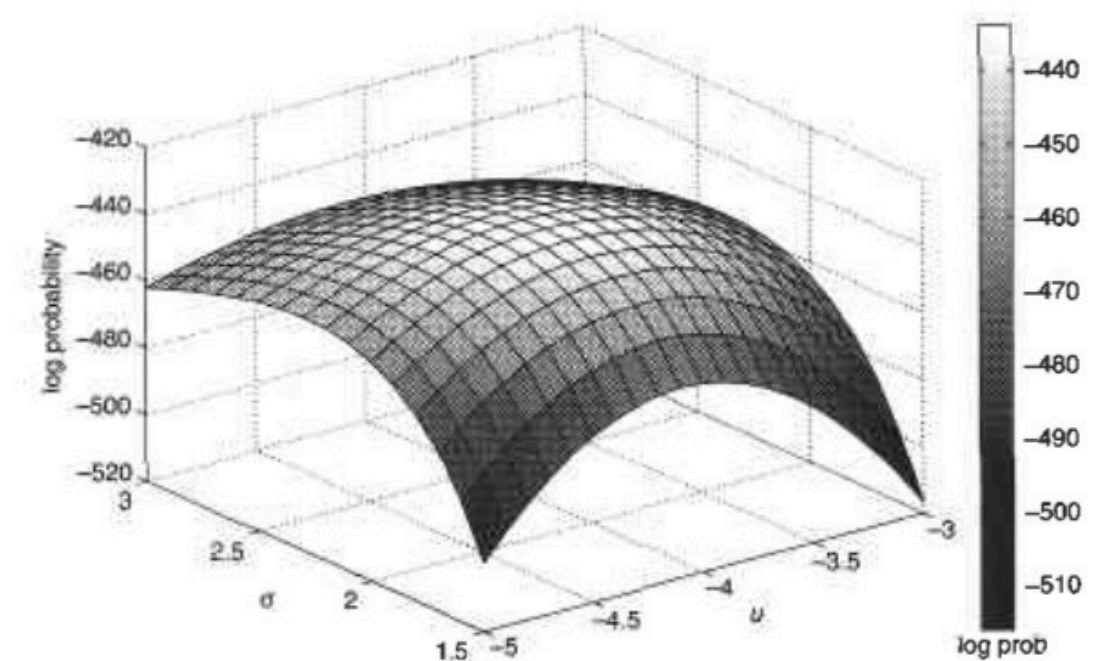
$$\log L(\theta|\mathbb{X}) = -\sum_{i=1}^n \frac{(x_i - \mu)^2}{2\sigma^2} - 0.5n \log(2\pi) - n \log(\sigma)$$

take log - nilu di $\in [0, \infty]$
- nilu sum ko 3 term
diff hu anumanavhu

Maximizing the log-likelihood function gives the parameters (μ, σ) best matched the data.



(a) Histogram of 200 points from a Gaussian distribution.

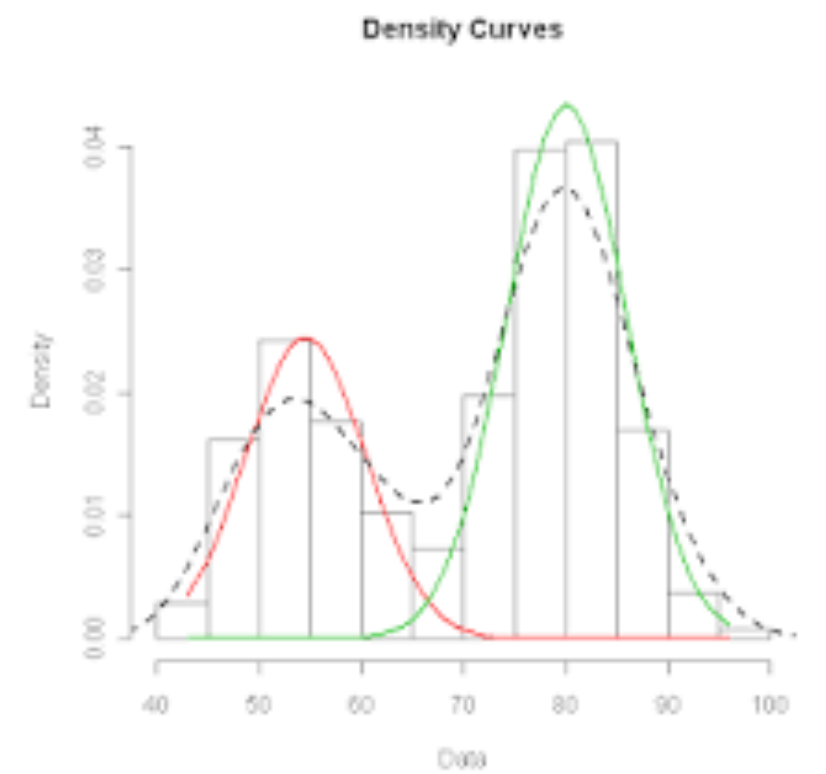


(b) Log likelihood plot of the 200 points for different values of the mean and standard deviation.

Gaussian Mixture Model (GMM) in 1D

Data as a set of observations from a weighted sum of K gaussian distributions $f_k(x; \theta)$.

$$f(x) = w_1 f_1(x) + w_2 f_2(x) + \dots + w_K f_K(x), \quad f_i(x) \sim \text{Normal}(\mu_i, \sigma_i)$$
$$f_i(x|\theta_i) = \frac{1}{\sqrt{2\pi}\sigma_i} e^{-\frac{(x-\mu_i)^2}{2\sigma_i^2}}, \quad \theta_i = (\mu_i, \sigma_i)$$



Each distribution describes a different group/cluster.

Each data point is probabilistically associated with all clusters in the clustering.

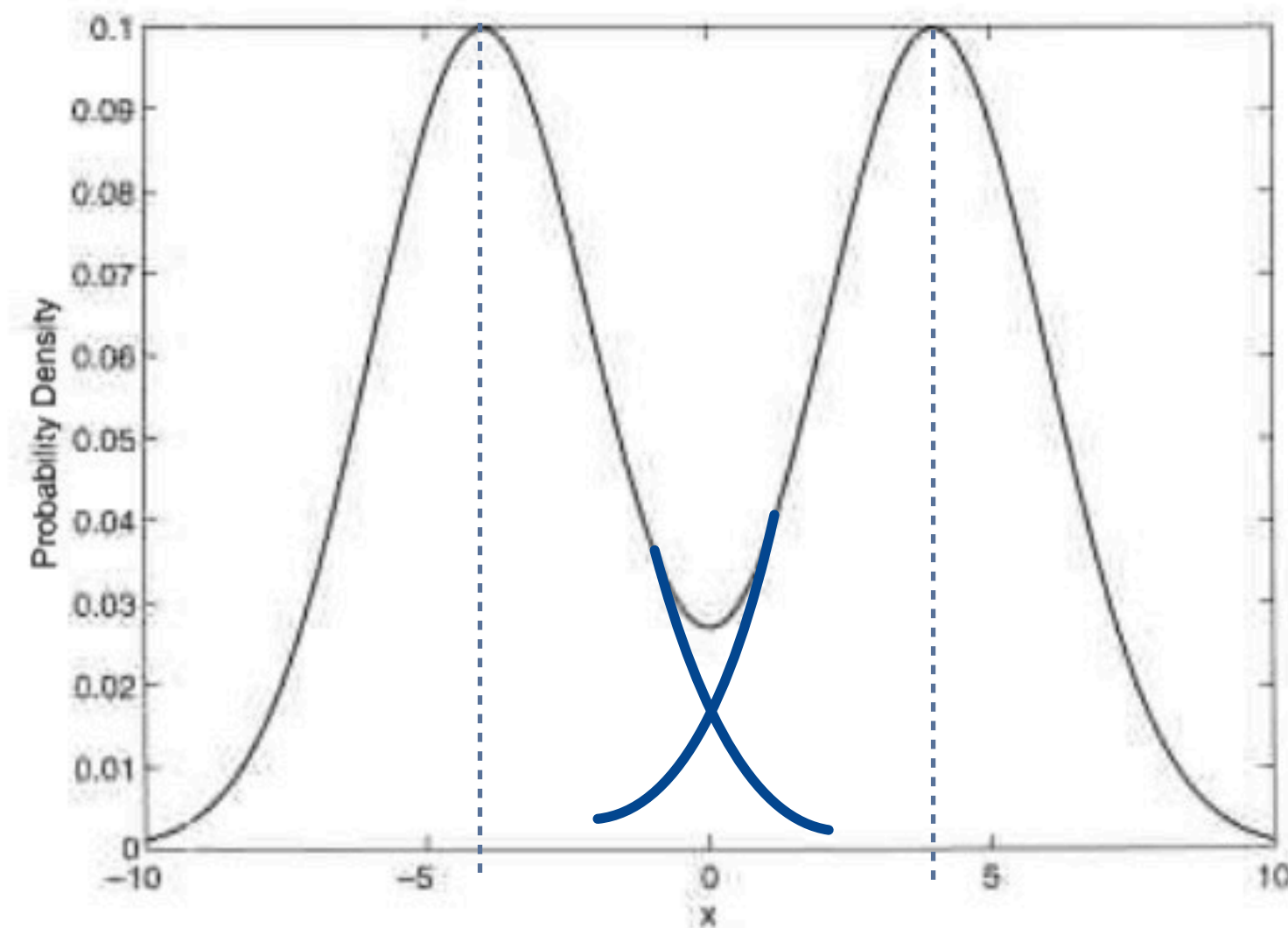
Consider a set of n observations $\mathbb{X} = \{x_1, x_2, \dots, x_n\}$ and two gaussian distributions with parameter set $\Theta = \{\theta_1, \theta_2\}$.

Then, for $w_1 + w_2 = 1$, and $0 < w_k < 1$, the probability distribution of x from the mixture model is

$$f(x|\Theta) = w_1 \overset{\text{cluster 1}}{f(x|\theta_1)} + w_2 \overset{\text{cluster 2}}{f(x|\theta_2)}$$

Ex: $w_1 = w_2 = 0.5$, $\theta_1 = (\mu_1, \sigma_1) = (-4, 2)$, $\theta_2 = (\mu_2, \sigma_2) = (4, 2)$,

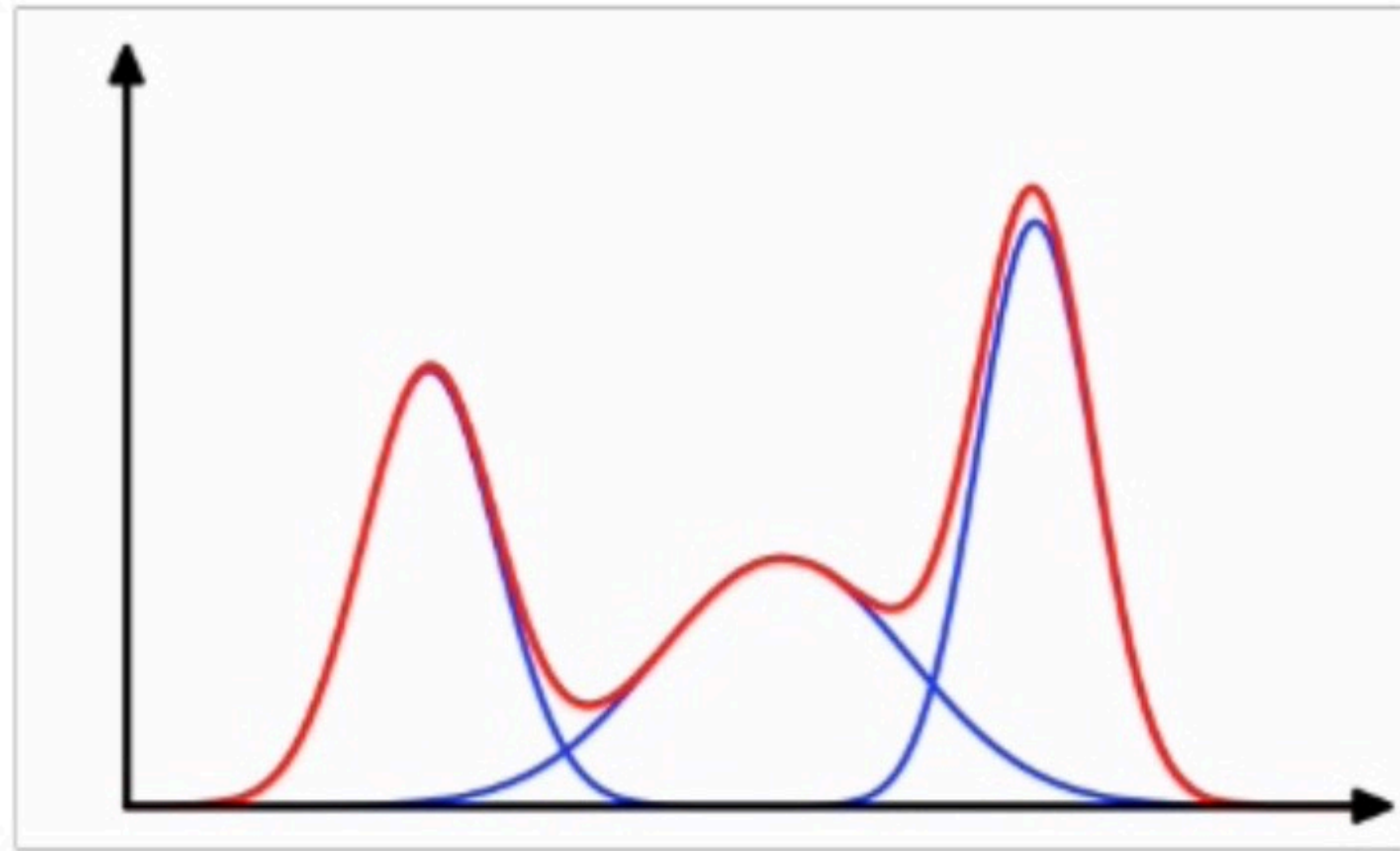
$$f(x|\Theta) = 0.5 \frac{1}{2\sqrt{2\pi}} e^{-\frac{(x+4)^2}{8}} + 0.5 \frac{1}{2\sqrt{2\pi}} e^{-\frac{(x-4)^2}{8}}$$



For $K = 3$ clusters,

$$f(x|\Theta) = w_1 f_1(x|\theta_1) + w_2 f_2(x|\theta_2) + w_3 f_3(x|\theta_3)$$

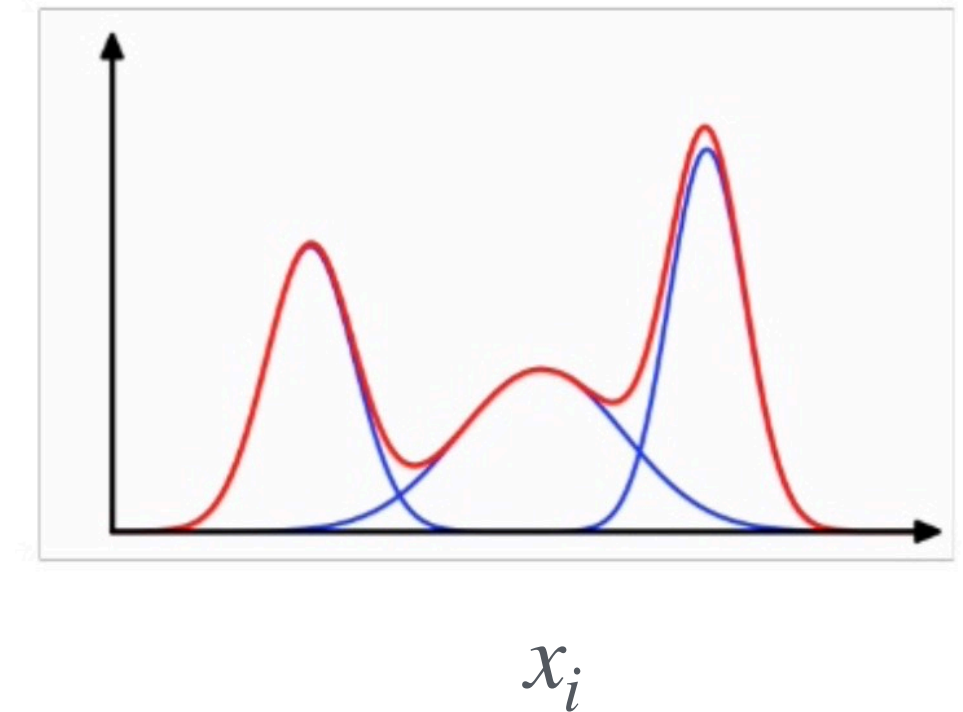
$$f_i(x|\theta_i) = \frac{1}{\sigma_i \sqrt{2\pi}} e^{-\frac{1}{2} \left(\frac{x - \mu_i}{\sigma_i} \right)^2}$$



Estimating Model Parameters in 1D Mixture Model

Given a data set $\mathbb{X} = \{x_1, x_2, \dots, x_n\}$, the likelihood function from a mixture of $K = 2$ gaussian distributions is

$$\begin{aligned} L(\Theta, \boldsymbol{w} | \mathbb{X}) &= \prod_{i=1}^n f(x_i | \Theta) \\ &= \prod_{i=1}^n \sum_{k=1}^K w_k f(x_i; \theta_k) \\ &= \prod_{i=1}^n \left[w_1 \frac{1}{\sqrt{2\pi}\sigma_1} e^{-\frac{(x_i - \mu_1)^2}{2\sigma_1^2}} + w_2 \frac{1}{\sqrt{2\pi}\sigma_2} e^{-\frac{(x_i - \mu_2)^2}{2\sigma_2^2}} \right] \end{aligned}$$



No closed-form solution for MLE unless w_i 's are known.

The model fitting maximizes the log likelihood function (LLF) by iteratively updating the model parameters.

Expectation Maximization (EM) Algorithm in 1D

Define a discrete variable $Z \in \{1, 2, \dots, K\}$ where $w_k \triangleq P\{Z = k\}$ or $P\{Z_k\}$ is the probability that a randomly selected data point comes from distribution/cluster k .

Given $\Theta = \{\theta_1, \theta_2\} = \{(\mu_1, \sigma_1), (\mu_2, \sigma_2)\}$ of two distributions in the mixture with weights w_1, w_2 , each data point x_i belongs to cluster k with probability

$$\begin{aligned} w_{ki} = P\{Z = k | x_i\} &= \frac{P\{Z_k, x_i\}}{P\{x_i\}}, \quad k = 1, 2 \\ &= \frac{P\{x_i | Z_k\} P\{Z_k\}}{\sum_j P\{x_i | Z_j\} P\{Z_j\}}, \quad (\text{Bayes' theorem}) \\ &\approx \frac{w_k f(x_i | \theta_k)}{\sum_j w_j f(x_i | \theta_j)} \end{aligned}$$

Ex: With initial values of $\Theta = \{\theta_1, \theta_2\}$ and $w_1 = w_2 = 0.5$, we have

$$\begin{aligned} w_{ki} &\approx \frac{0.5 f(x_i | \theta_k)}{0.5 f(x_i | \theta_1) + 0.5 f(x_i | \theta_2)} \\ \hat{w}_k &\leftarrow (1/n) \sum_i w_{ki} \quad (\text{Expectation Step}) \end{aligned}$$

Plugging the estimated values of w_1, w_2 into the likelihood function:

$$L(\Theta | \hat{\mathbf{w}}, \mathbb{X}) = \prod_{i=1}^n \left[\hat{w}_1 \frac{1}{\sqrt{2\pi}\sigma_1} e^{-\frac{(x_i - \mu_1)^2}{2\sigma_1^2}} + \hat{w}_2 \frac{1}{\sqrt{2\pi}\sigma_2} e^{-\frac{(x_i - \mu_2)^2}{2\sigma_2^2}} \right]$$

It turns out that $L(\Theta | \hat{\mathbf{w}}, \mathbb{X})$ is maximized at

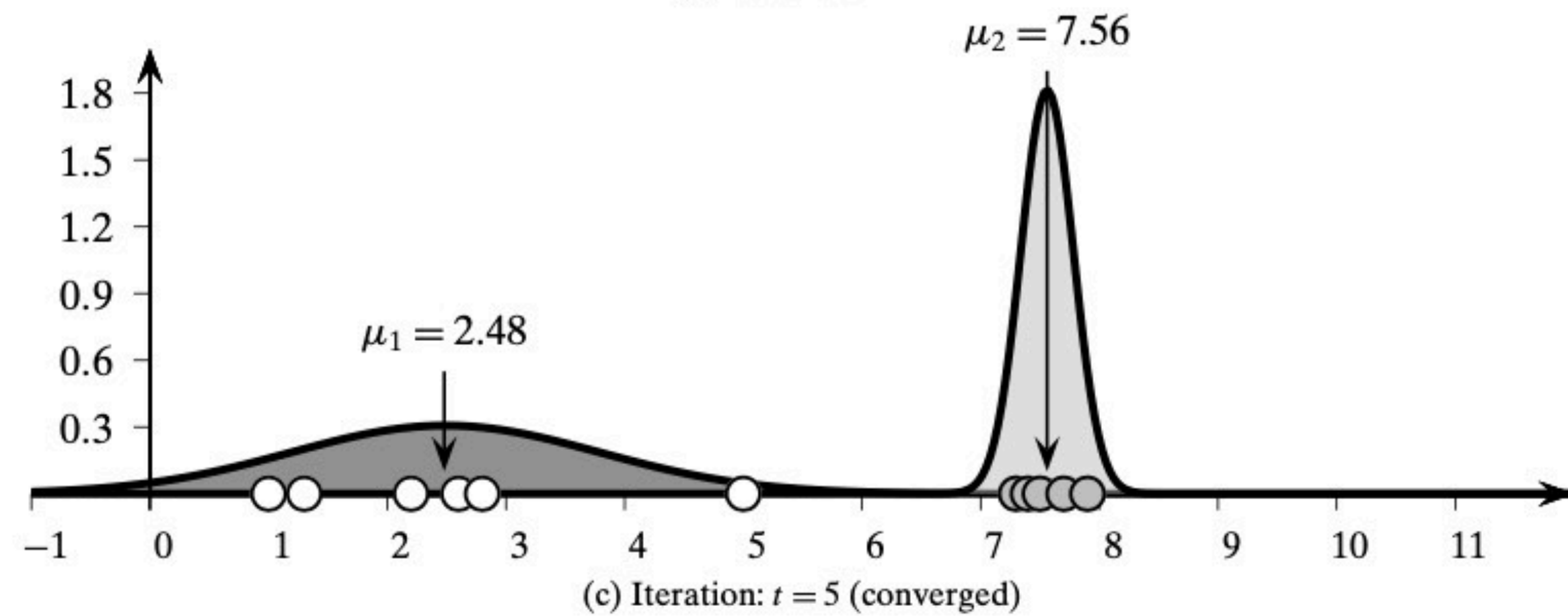
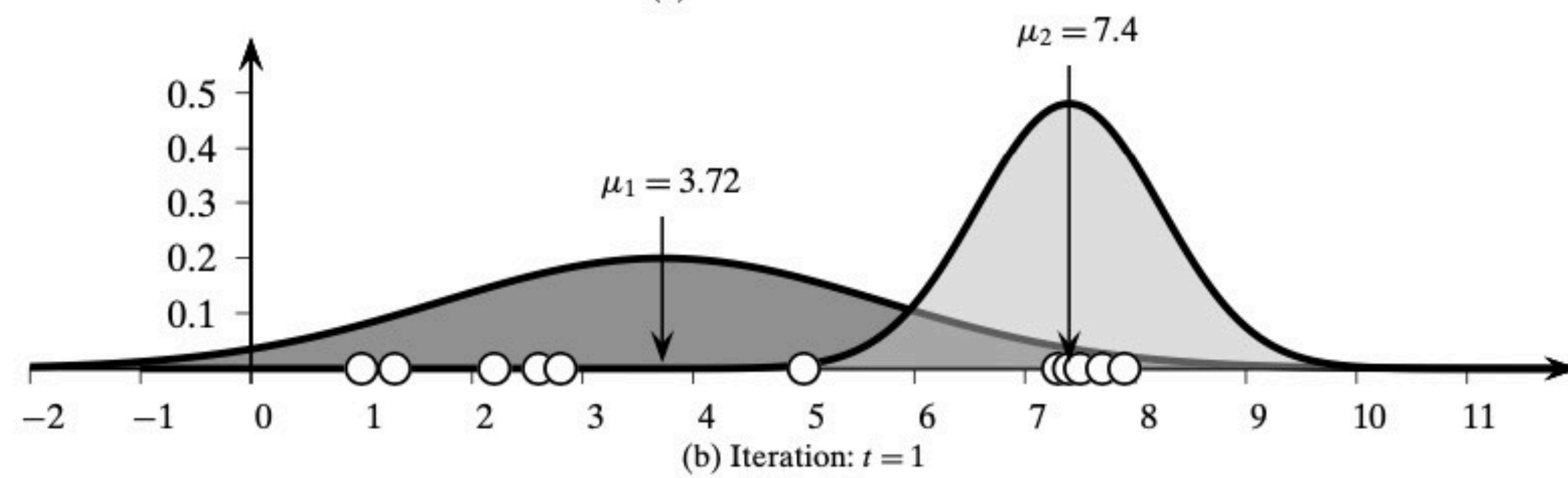
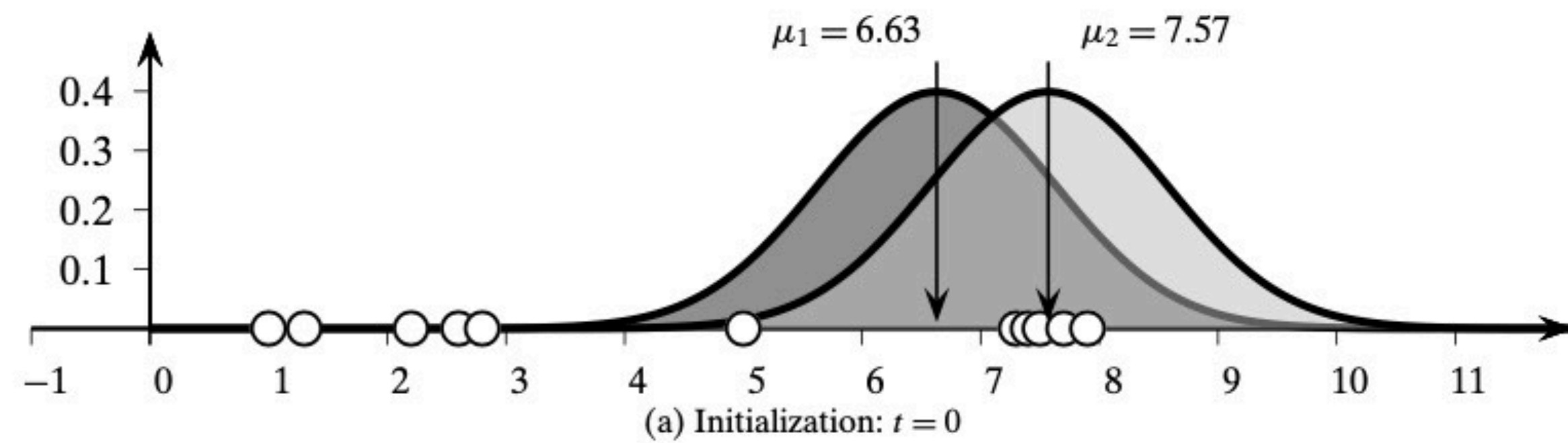
$$\left. \begin{aligned} \hat{\mu}_k &= \frac{1}{n} \sum_i \frac{w_{ki}}{\hat{w}_k} x_i \\ \hat{\sigma}_k^2 &= \frac{1}{n} \sum_i \frac{w_{ki}}{\hat{w}_k} (x_i - \hat{\mu}_k)^2 \end{aligned} \right\} \quad (\text{Maximization Step})$$

Substituting the updated parameters $\hat{\mu}_k, \hat{\sigma}_k^2$ to $f(x | \theta_i)$, recompute $\hat{w}_{ki}$, and repeat until $\hat{\mu}_k$ converges.

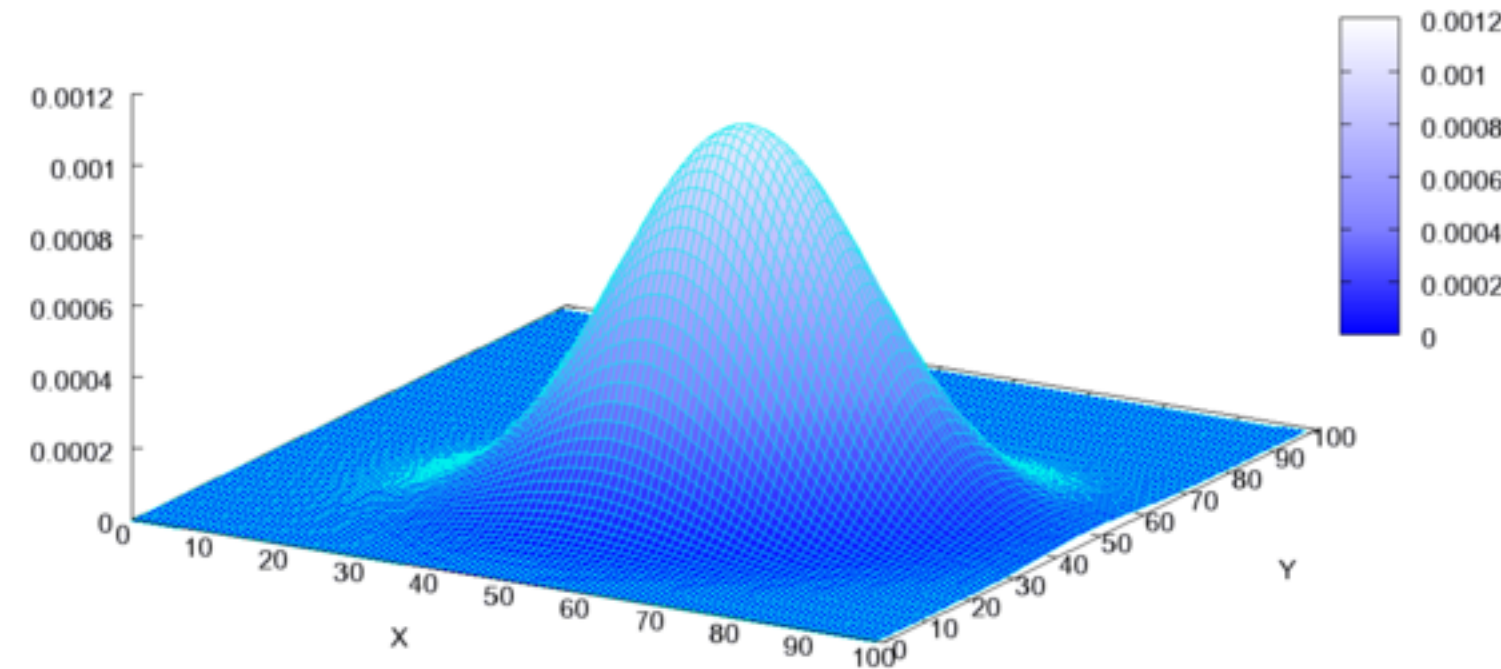
How to determine the cluster membership for a given data point ?

Algorithm (Univariate) Gaussian Mixture EM Algorithm

- 1: **Input:** n data points $\{x_1, x_2, \dots, x_n\}$ and the number of clusters K
 - 2: **Output:** K model parameters $\{(\mu_k, \sigma_k)\}$, $1 \leq k \leq K$
 - 3:
 - 4: **Initialization**
 - 5: Initialize $\hat{\mu}_k$
 - 6: $\hat{\sigma}_k^2 \leftarrow \frac{1}{n} \sum_i (x_i - \bar{x})^2$
 - 7: $w_k \leftarrow 1/K$
 - 8:
 - 9: **repeat**
 - 10: **Expectation Step**
 - 11: Compute for all data points x_i , $w_{ki} \leftarrow \frac{w_k f(x_i | \theta_k)}{\sum_{j=1}^K w_j f(x_i | \theta_j)}$, $\forall k$
 - 12: $w_k \leftarrow (1/n) \sum_i w_{ki}$, $\forall k$
 - 13:
 - 14: **Maximization Step**
 - 15: $\hat{\mu}_k \leftarrow \frac{1}{n} \sum_i \frac{w_{ki}}{w_k} x_i$
 - 16: $\hat{\sigma}_k^2 \leftarrow \frac{1}{n} \sum_i \frac{w_{ki}}{w_k} (x_i - \hat{\mu}_k)^2$
 - 17: **until** $\hat{\mu}_k$ converges
 - 18: **return**
-



Multivariate Normal Distribution



$$\mathbf{X} = (X_1, X_2, \dots, X_d)$$

$$f(\mathbf{x}|\boldsymbol{\theta}) = \frac{\exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu})\right)}{\sqrt{(2\pi)^d |\boldsymbol{\Sigma}|}}, \quad \mathbf{x}, \boldsymbol{\mu} \in \mathbb{R}^d, \boldsymbol{\Sigma} \in \mathbb{R}^{d \times d}$$

Parameters : $\boldsymbol{\theta} = (\boldsymbol{\mu}, \boldsymbol{\Sigma})$

For $d = 2$: $\boldsymbol{\mu} = \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix}, \quad \boldsymbol{\Sigma} = \begin{bmatrix} \sigma_1^2 & \rho_{12}\sigma_1\sigma_2 \\ \rho_{12}\sigma_1\sigma_2 & \sigma_2^2 \end{bmatrix}$

$$|\boldsymbol{\Sigma}| = \sigma_1^2 \sigma_2^2 (1 - \rho_{12}^2)$$

$$f(x_1, x_2|\boldsymbol{\theta}) = \frac{\exp\left\{-\frac{1}{2(1-\rho^2)}\left[\left(\frac{x_1-\mu_1}{\sigma_1}\right)^2 - 2\rho\left(\frac{x_1-\mu_1}{\sigma_1}\right)\left(\frac{x_2-\mu_2}{\sigma_2}\right) + \left(\frac{x_2-\mu_2}{\sigma_2}\right)^2\right]\right\}}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}}$$

Assume a set of n observations $\mathbb{X} = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n\}$, $\mathbf{x}_i \in \mathbb{R}^d$ and K multivariate normal distributions with parameter set $\Theta = (\boldsymbol{\theta}_1, \boldsymbol{\theta}_2, \dots, \boldsymbol{\theta}_K)$,

Then, for $\sum_i w_k = 1$, and $0 < w_k < 1$, the probability distribution of $\mathbf{x}$ given the mixture model is

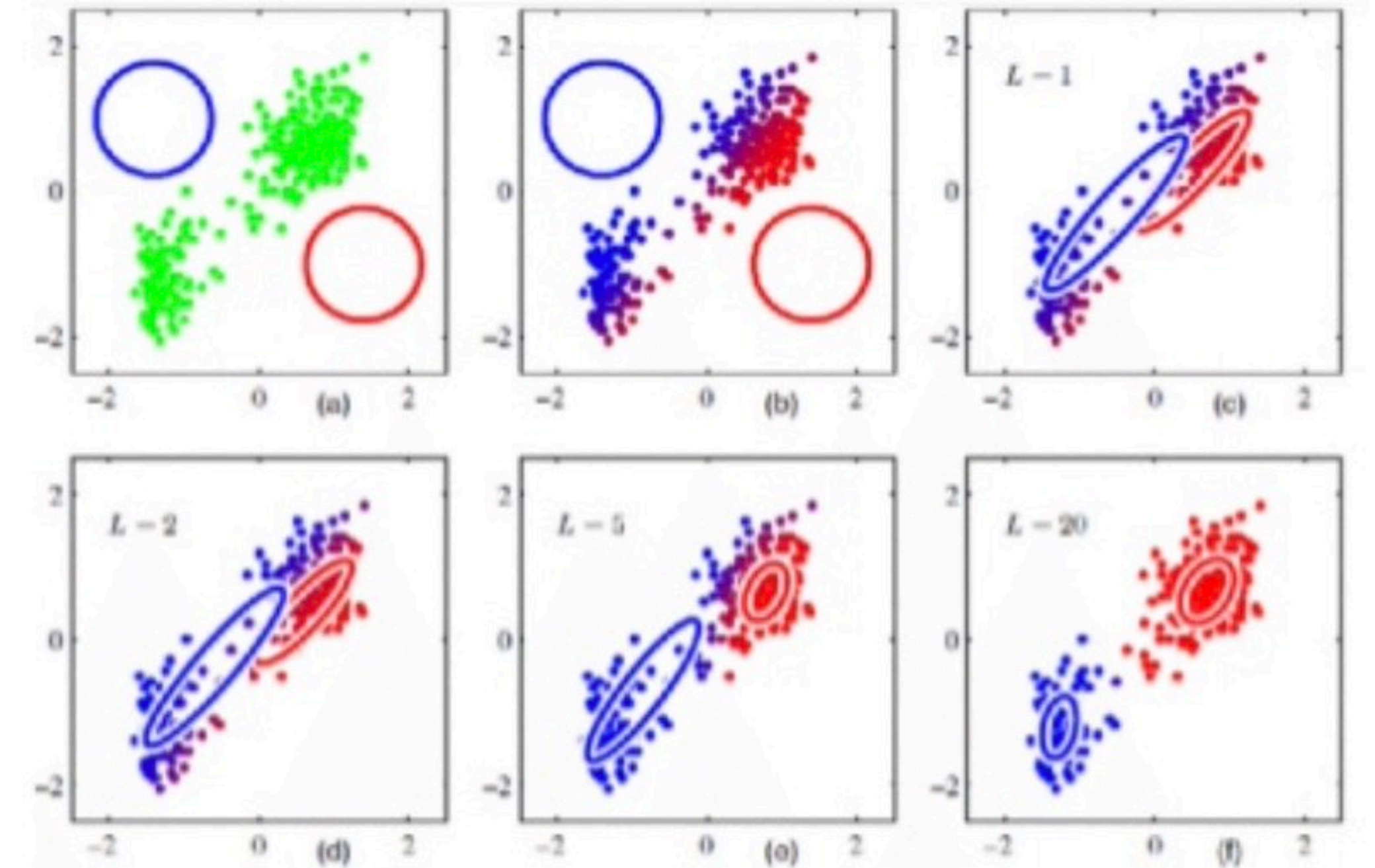
$$f(\mathbf{x}|\Theta) = \sum_{k=1}^K w_k f(\mathbf{x}|\boldsymbol{\theta}_k)$$

Given a data set $\mathbb{X} = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n\}$, the likelihood function is

$$\begin{aligned} L(\Theta, \boldsymbol{w}|\mathbb{X}) &= \prod_{i=1}^n f(\mathbf{x}_i|\Theta) \\ &= \prod_{i=1}^n \sum_{k=1}^K w_k f(\mathbf{x}_i; \boldsymbol{\theta}_k) \end{aligned}$$

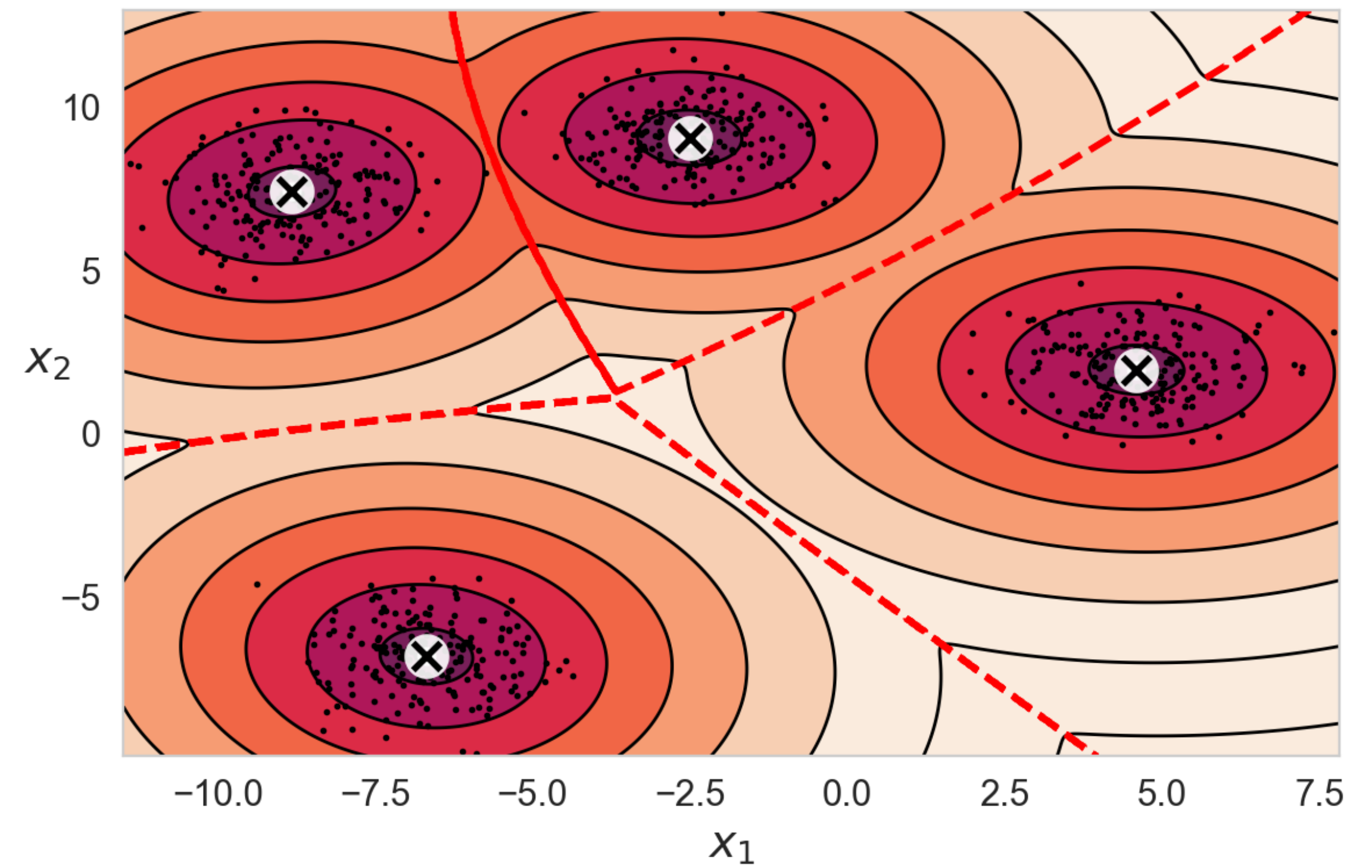
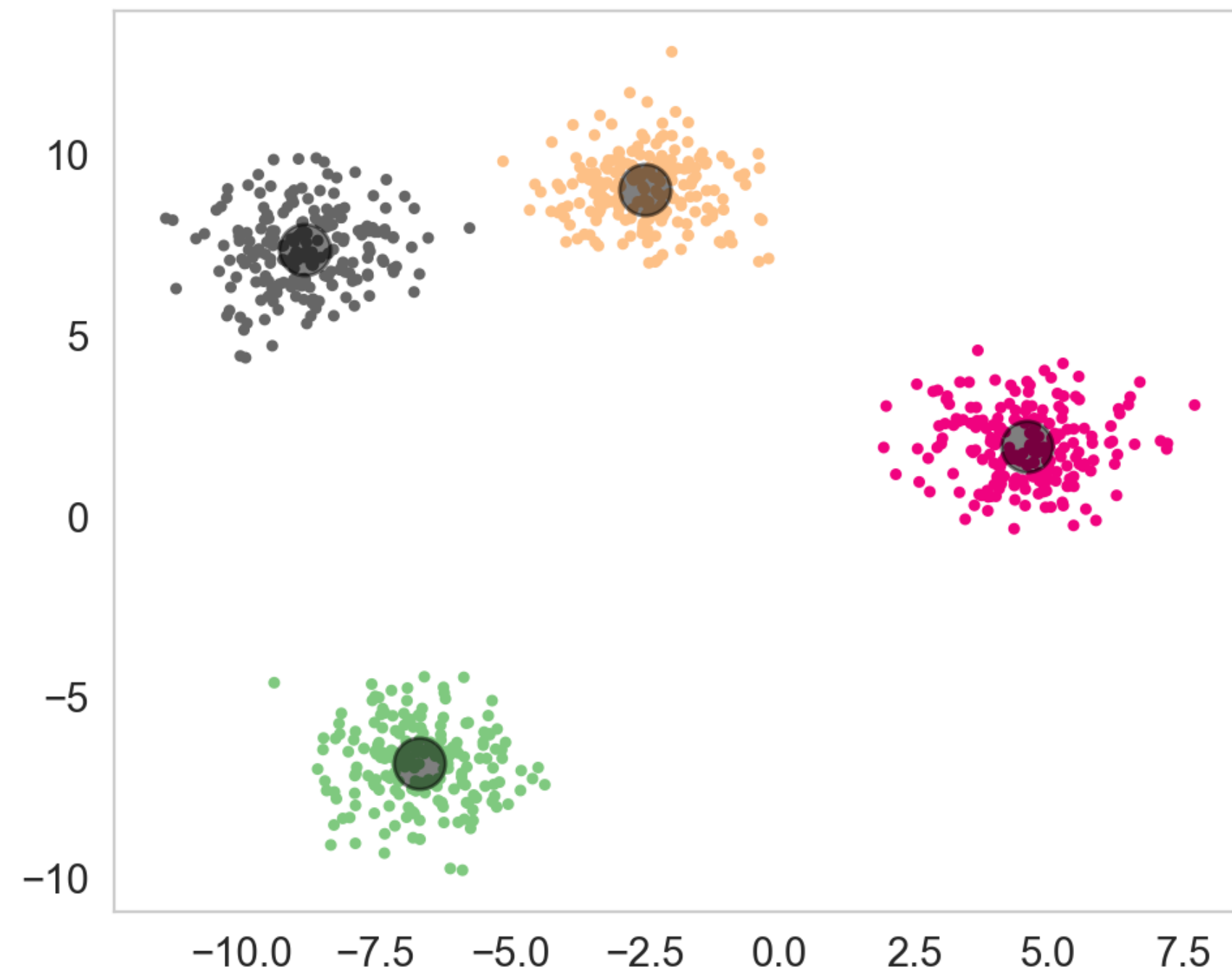
Algorithm (Multivariate) Gaussian Mixture EM Algorithm

- 1: **Input:** n data points $\mathbf{X} = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n\}$, $\mathbf{x}_i \in \mathbb{R}^d$, $\mathbf{X} \in \mathbb{R}^{d \times n}$, and the number of clusters K
 - 2: **Output:** K model parameters $\{(\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)\}$, $1 \leq k \leq K$, $\boldsymbol{\mu}_k \in \mathbb{R}^d$, $\boldsymbol{\Sigma}_k \in \mathbb{R}^{d \times d}$
 - 3:
 - 4: **Initialization**
 - 5: Initialize $\hat{\boldsymbol{\mu}}_k$ (e.g., random, K-means)
 - 6: $\hat{\boldsymbol{\Sigma}}_k \leftarrow \frac{1}{n} (\mathbf{X} - \bar{\mathbf{x}})(\mathbf{X} - \bar{\mathbf{x}})^T$
 - 7: $w_k \leftarrow 1/K$
 - 8: $\boldsymbol{\theta}_k \leftarrow (\hat{\boldsymbol{\mu}}_k, \hat{\boldsymbol{\Sigma}}_k)$
 - 9:
 - 10: **repeat**
 - 11: **Expectation Step**
 - 12: Compute for all data points $\mathbf{x}_i$, $w_{ki} \leftarrow \frac{w_k f(\mathbf{x}_i | \boldsymbol{\theta}_k)}{\sum_{j=1}^K w_j f(\mathbf{x}_i | \boldsymbol{\theta}_j)}$
 - 13: $w_k \leftarrow (1/n) \sum_i w_{ki}$
 - 14:
 - 15: **Maximization Step**
 - 16: $\hat{\boldsymbol{\mu}}_k \leftarrow \frac{1}{n} \sum_i \frac{w_{ki}}{w_k} \mathbf{x}_i$
 - 17: $\hat{\boldsymbol{\Sigma}}_k \leftarrow \frac{1}{n} \frac{\sum_{i=1}^n w_{ki} (\mathbf{x}_i - \hat{\boldsymbol{\mu}}_k)(\mathbf{x}_i - \hat{\boldsymbol{\mu}}_k)^T}{w_k}$
 - 18: **until** $\hat{\boldsymbol{\mu}}_k$ converges
 - 19: **return**
-





```
from sklearn.mixture import GaussianMixture
gm = GaussianMixture(n_components=4, n_init=10, random_state=42)
gm.fit(X)
```



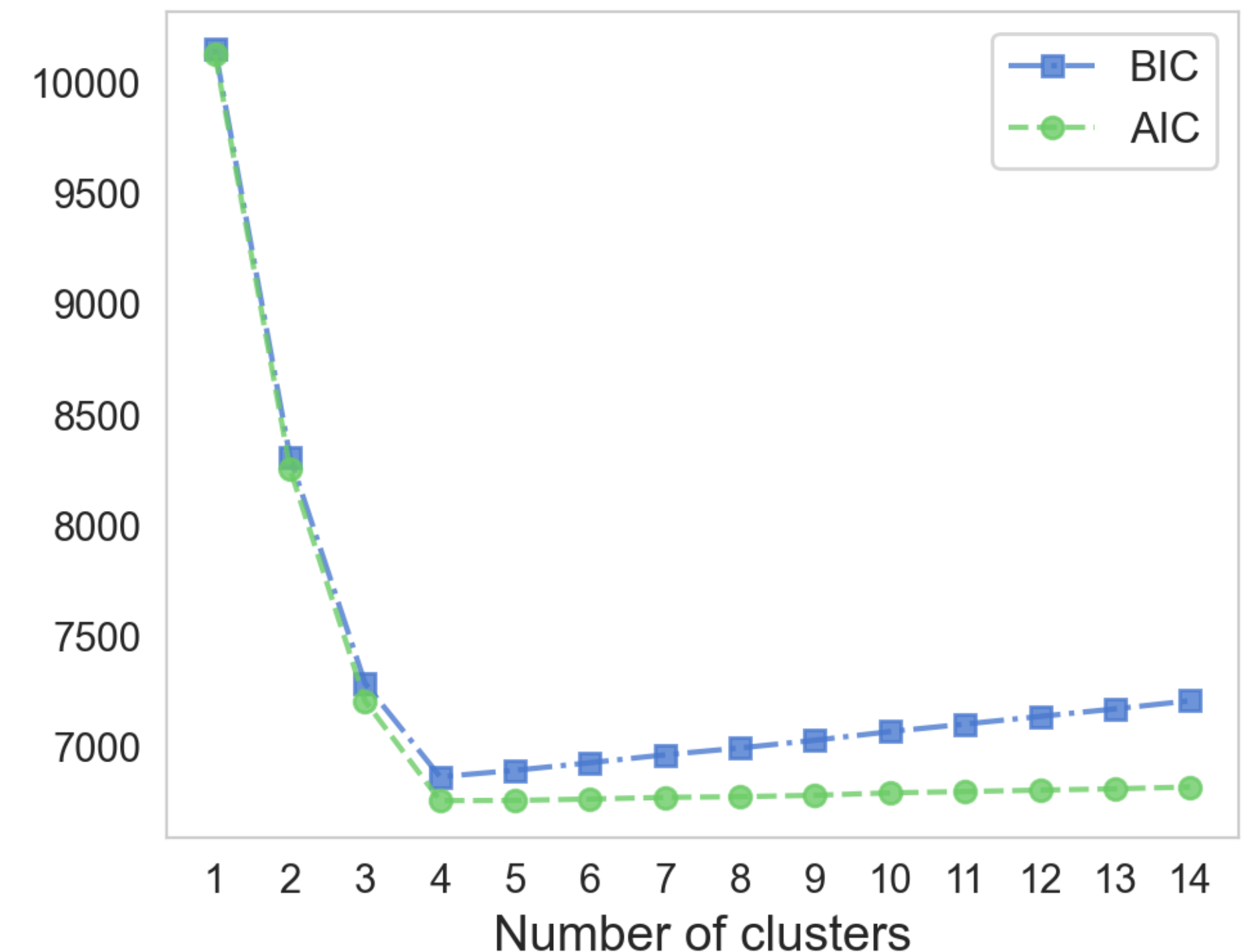
Determining the number of clusters in GMM

Common criteria to assess how well the model fits the data are AIC and BIC:

$$\text{AIC} = 2m - 2 \log(L)$$

$$\text{BIC} = m \log(N) - 2 \log(L)$$

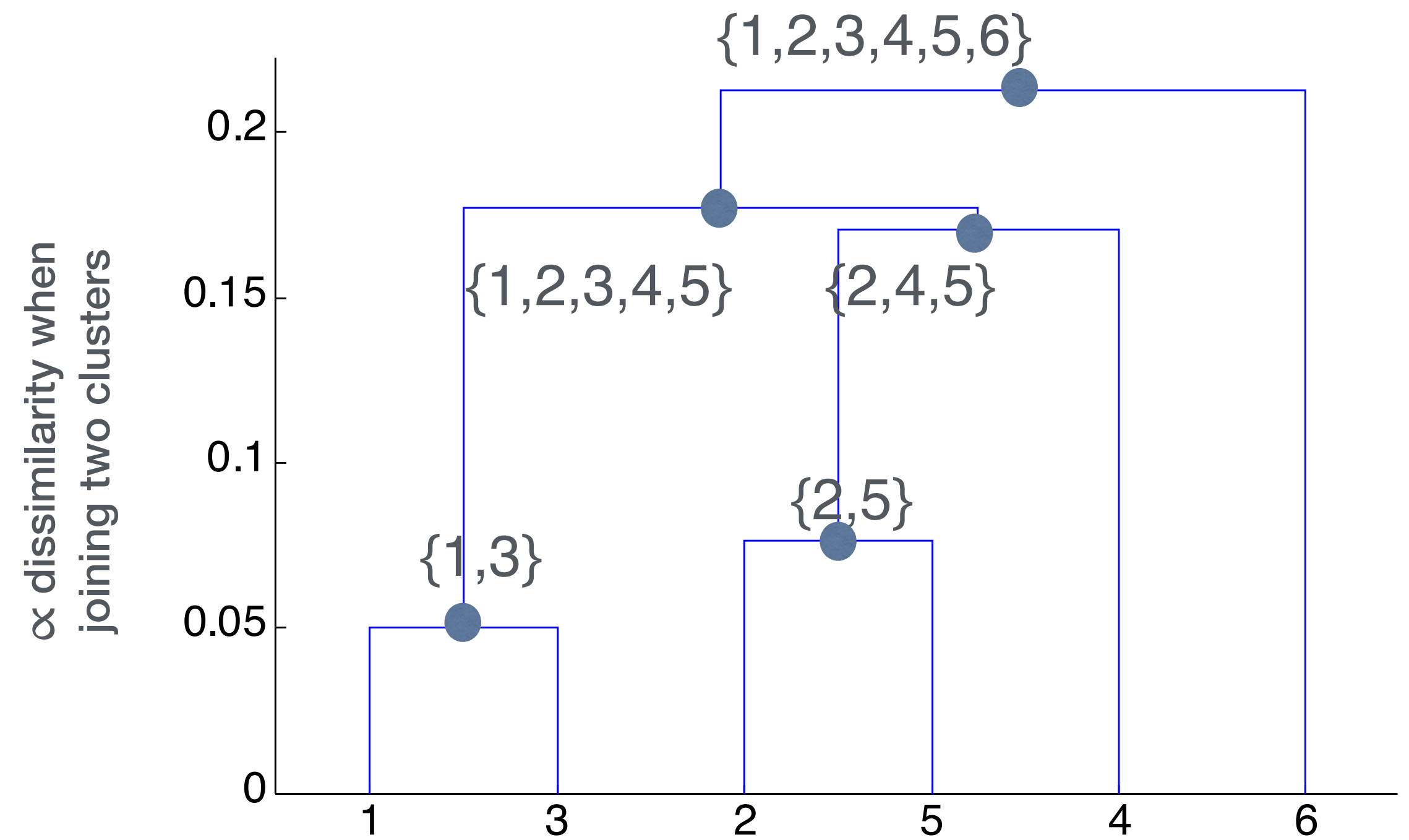
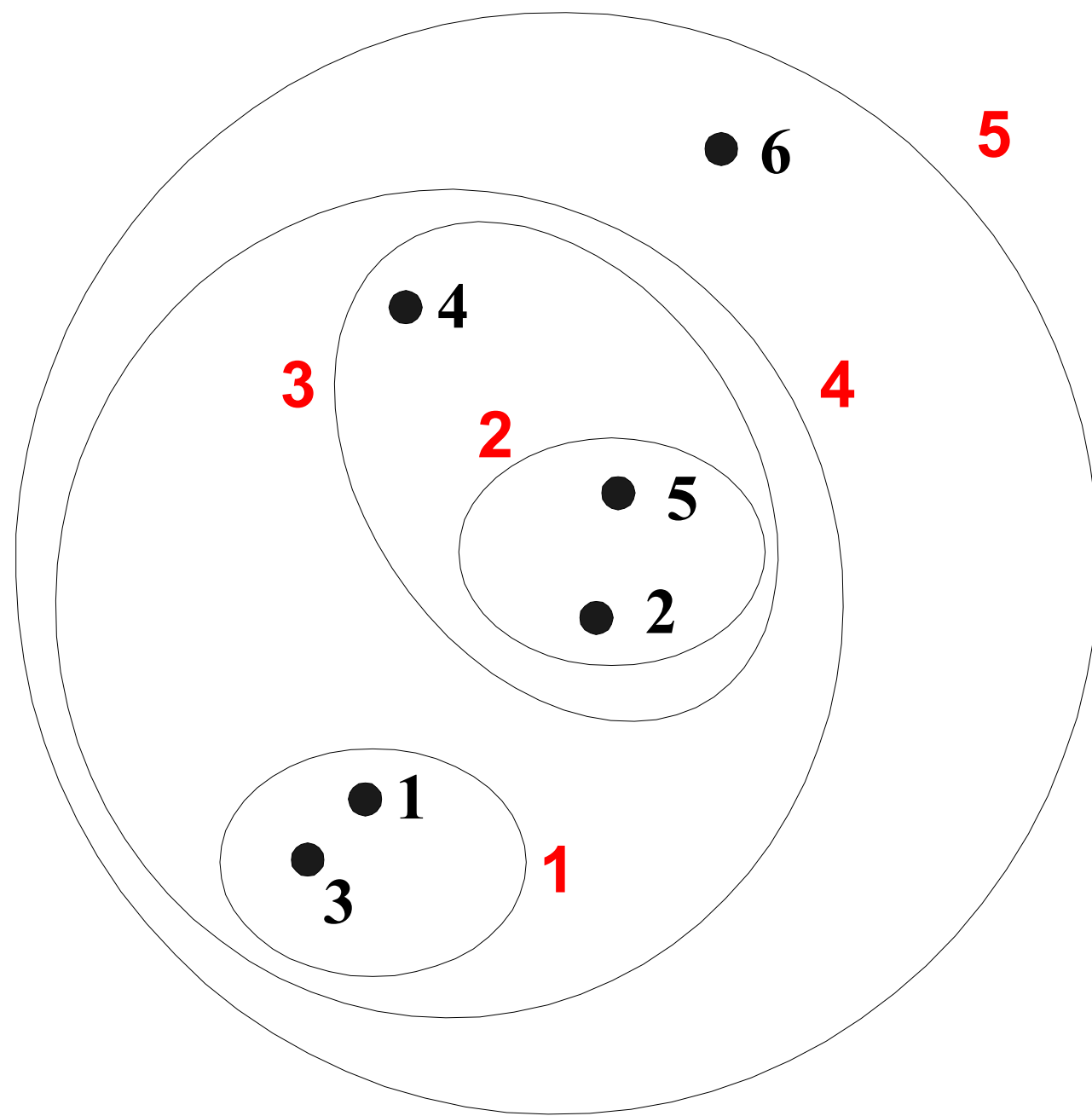
- ◆ L : Maximum likelihood value
- ◆ m : Number of model parameters
- ◆ N : Number of data points



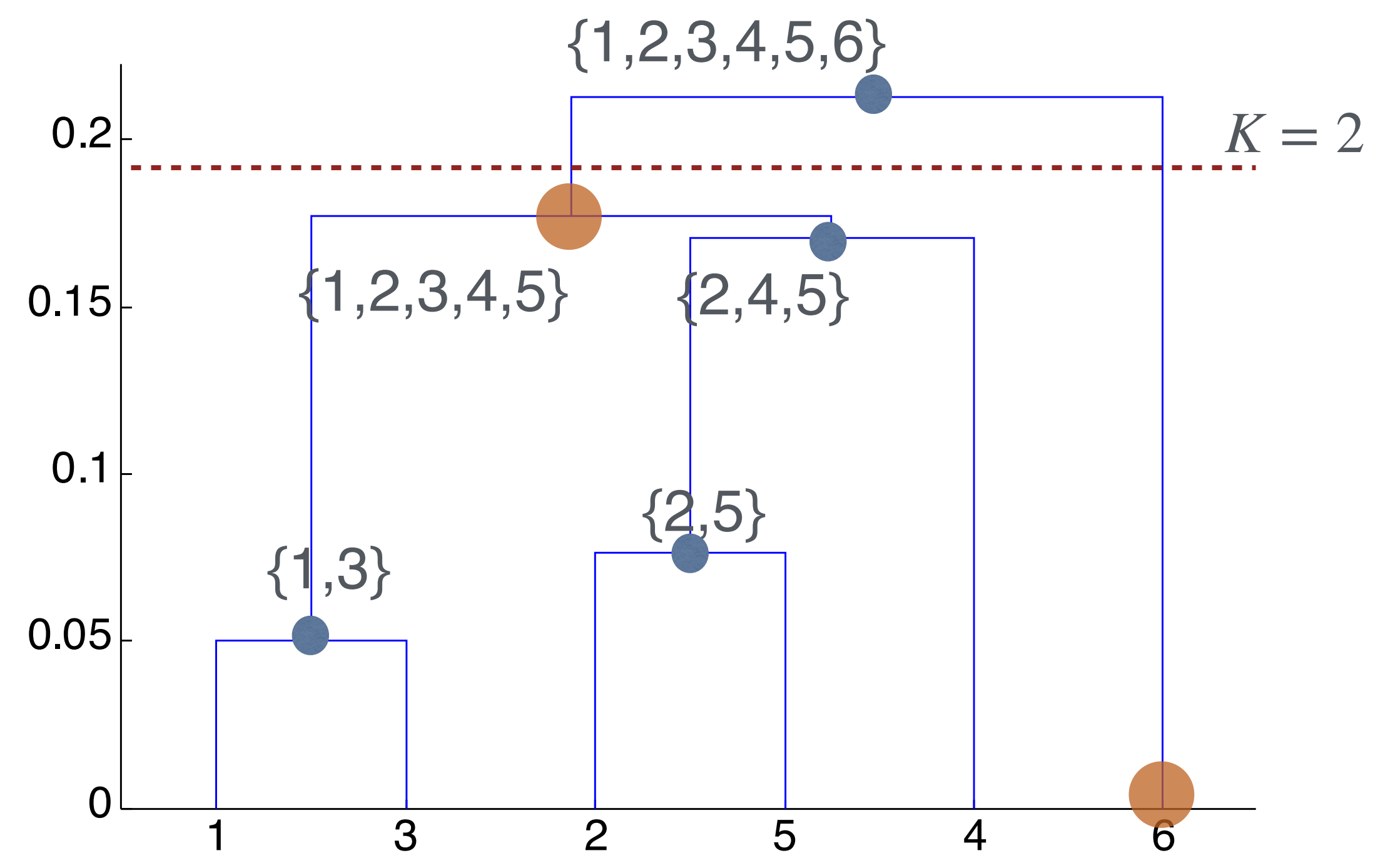
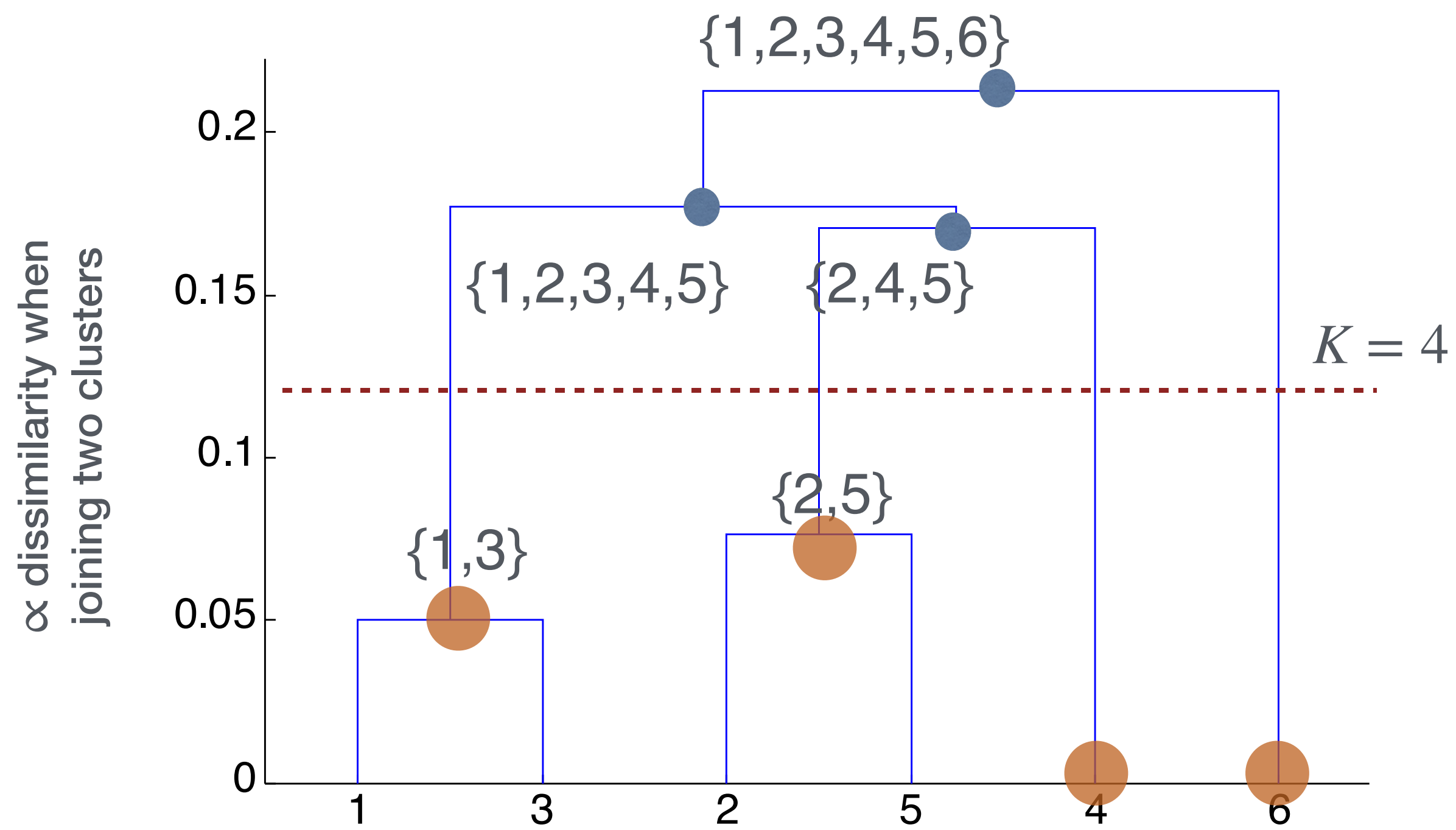
(Agglomerative) Hierarchical Clustering

Iteratively merge two *least dissimilar clusters* until getting one cluster containing all points.

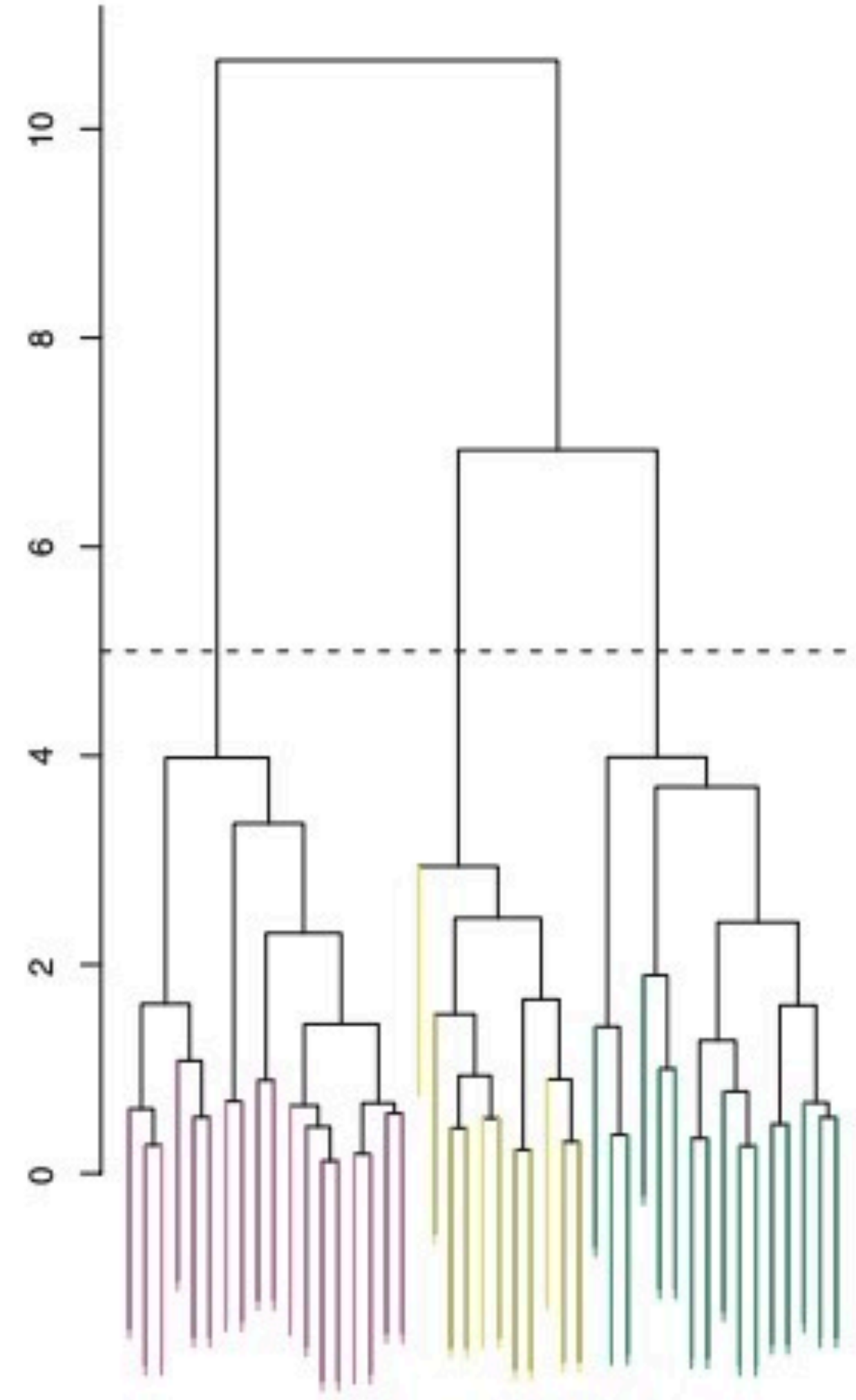
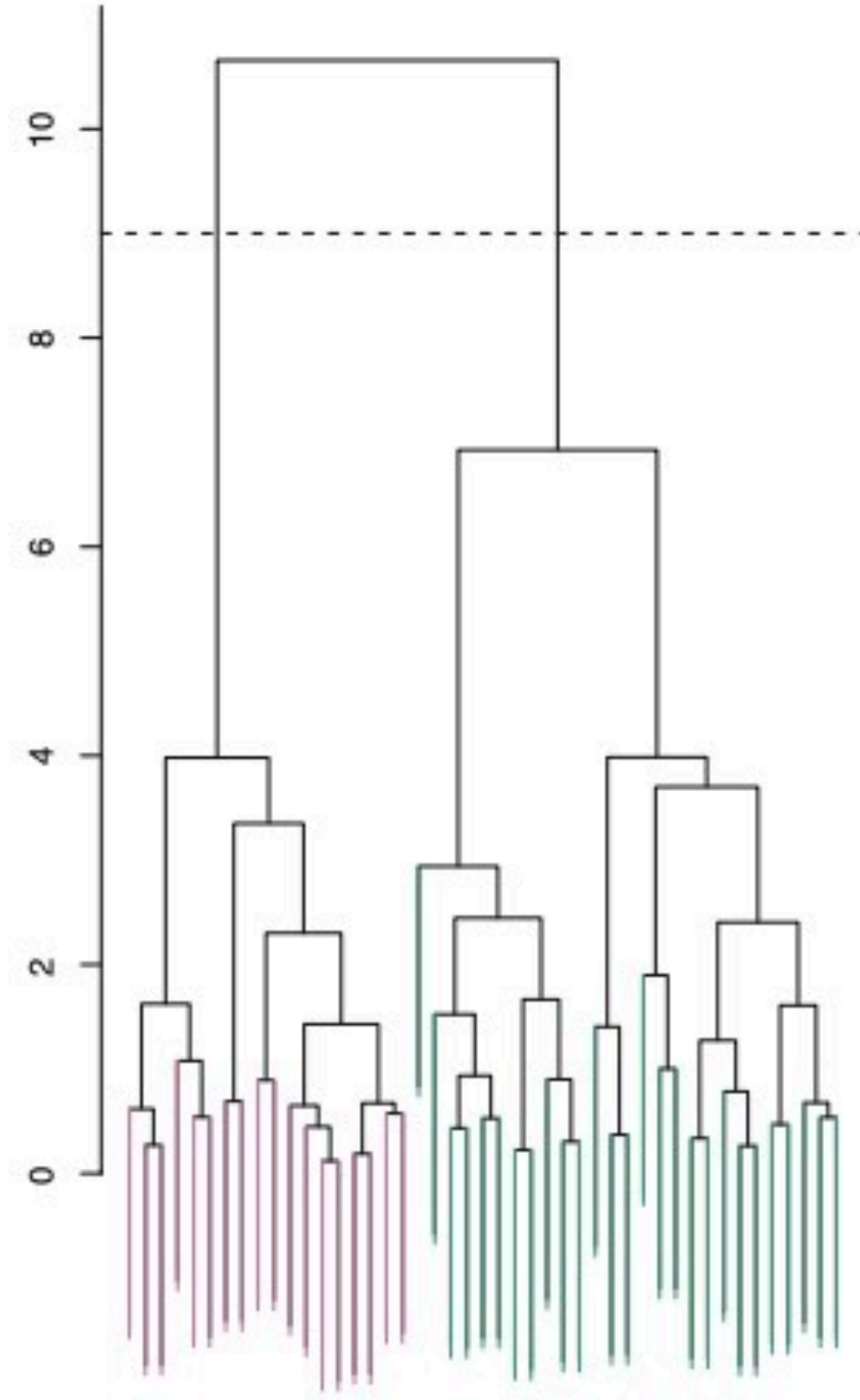
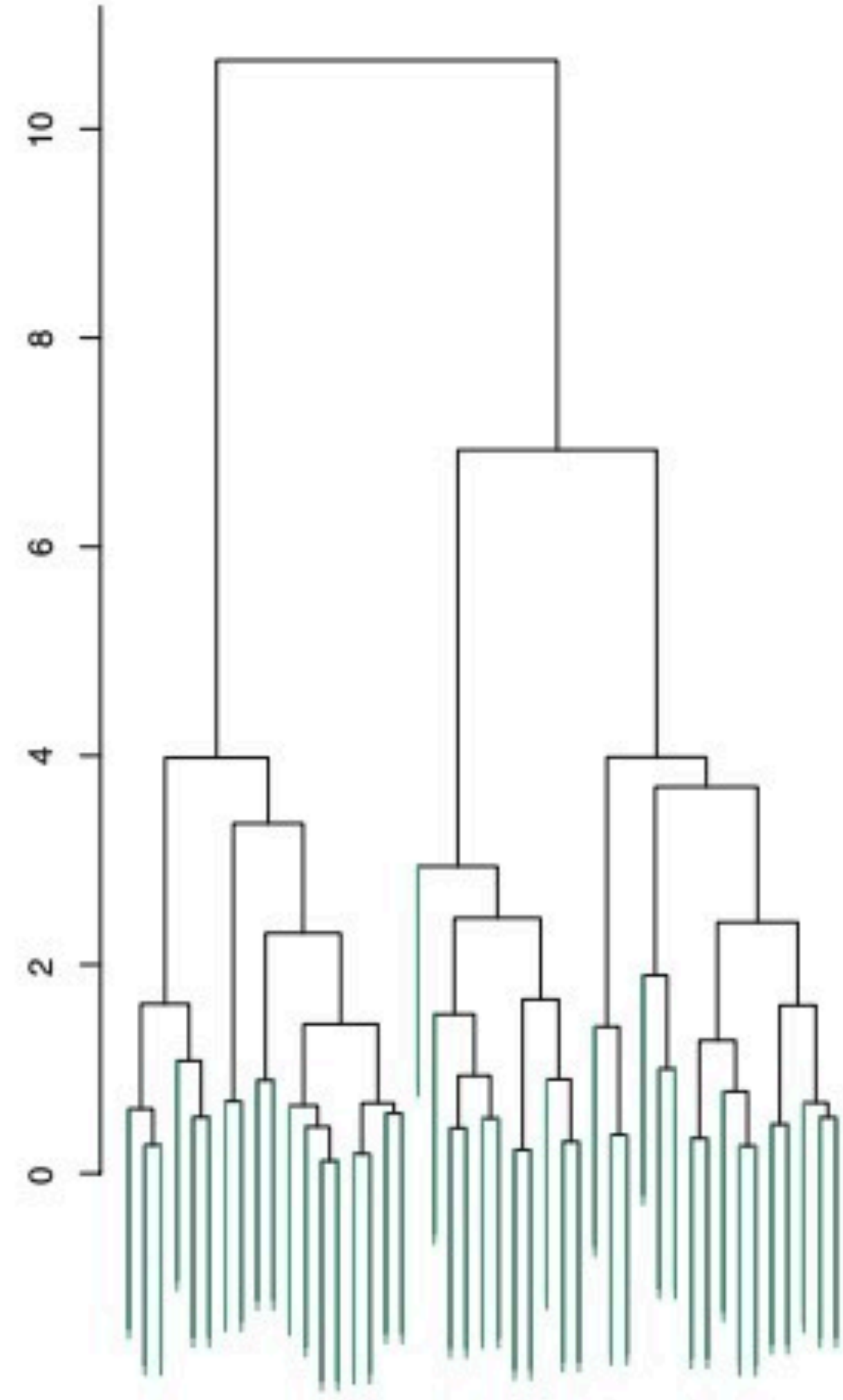
Dissimilarities of merged clusters monotonically increases with the level of the merger.



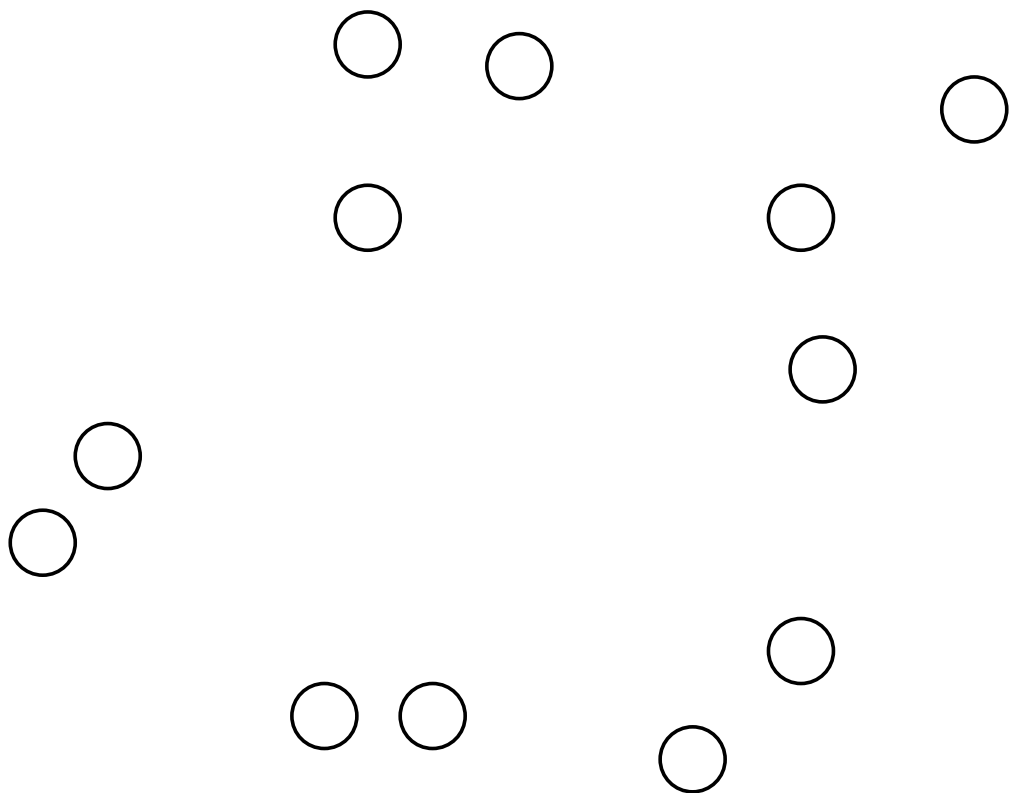
Dendrogram



Node height = Dissimilarity of two merged clusters

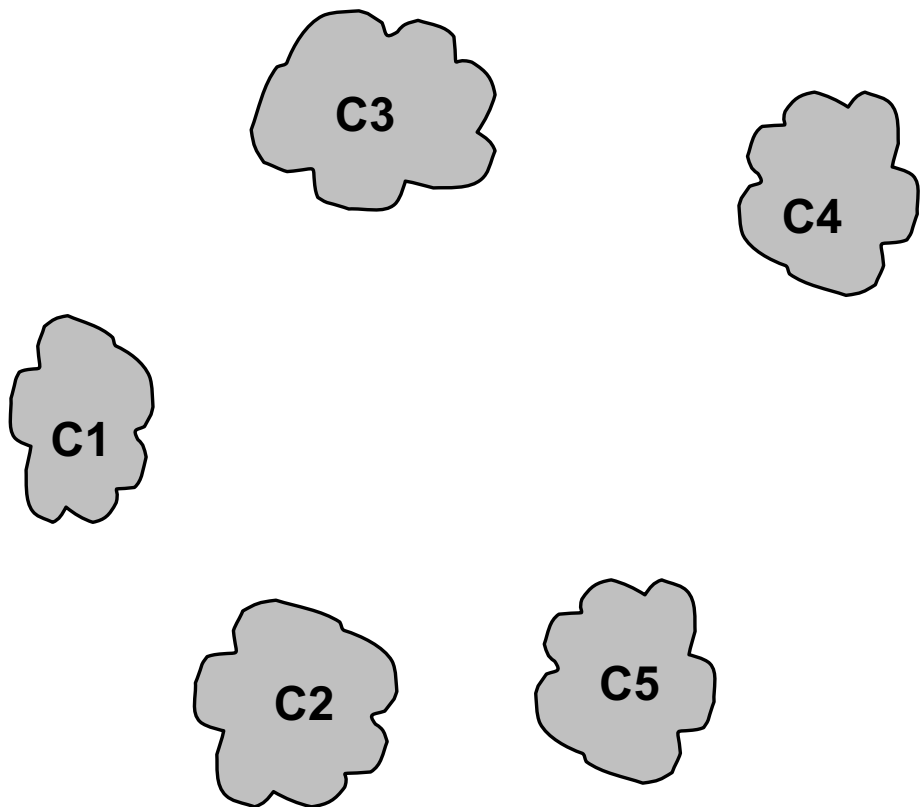
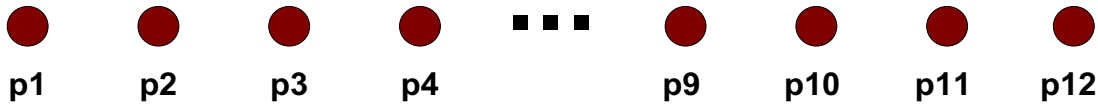


Merging Clusters



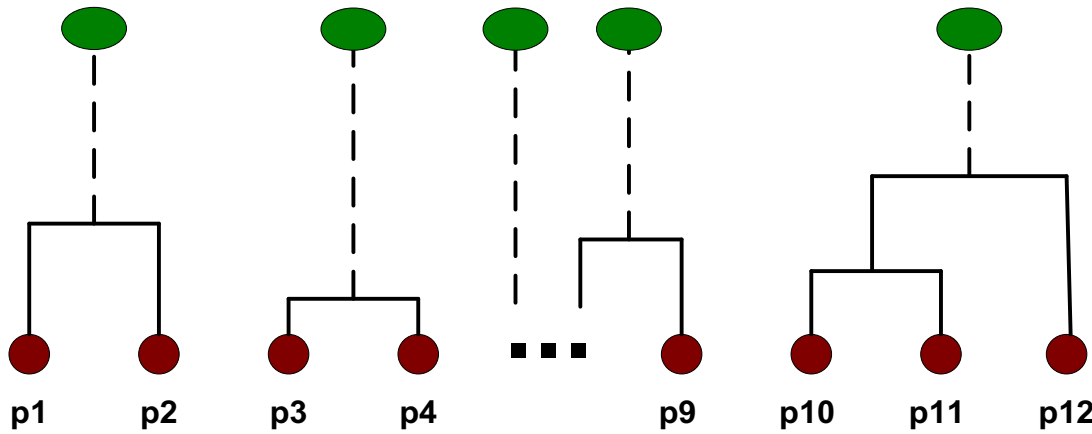
	p1	p2	p3	p4	p5	. . .
p1						
p2						
p3						
p4						
p5						
.						
.						
.						

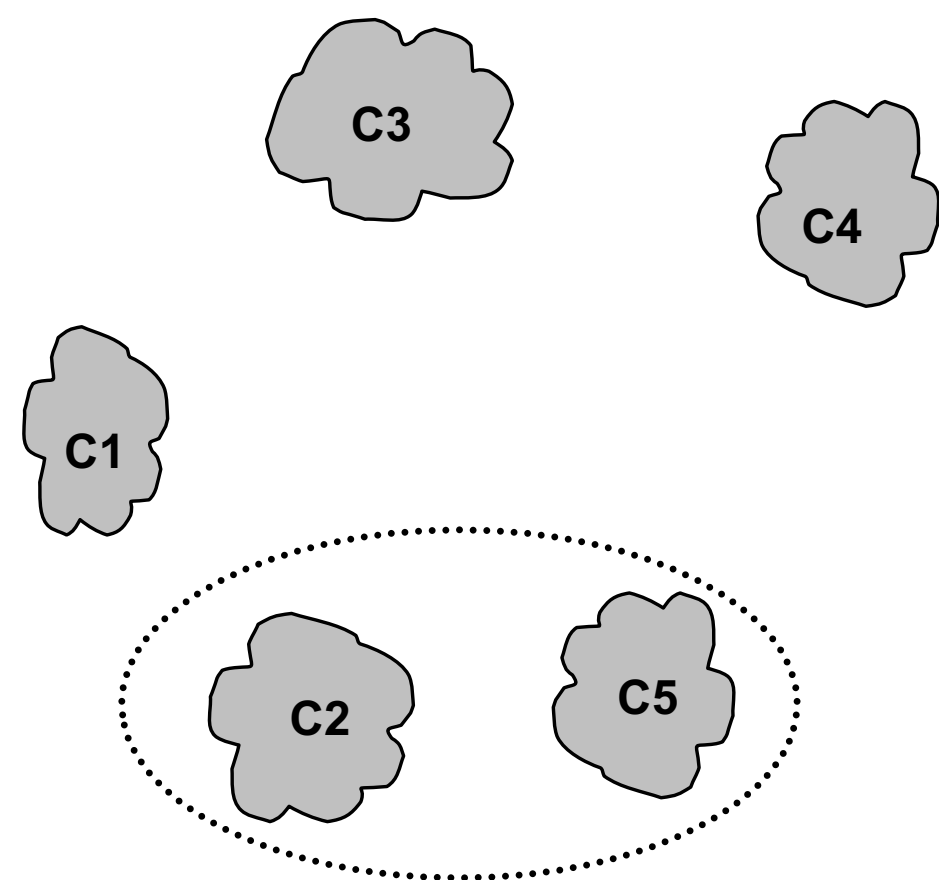
Proximity Matrix



	C1	C2	C3	C4	C5
C1					
C2					
C3					
C4					
C5					

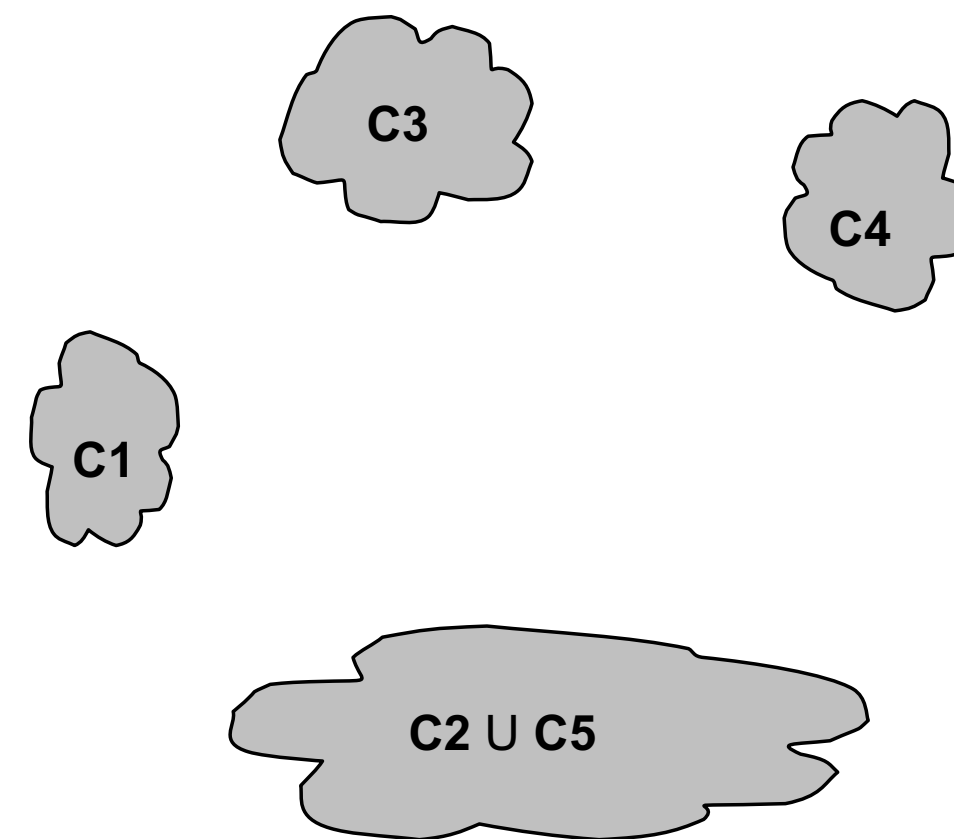
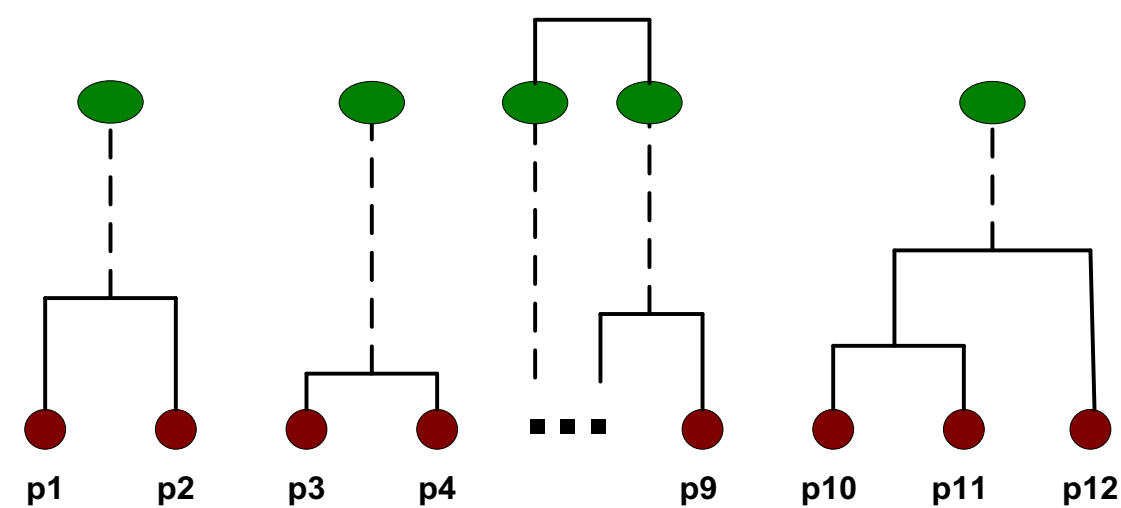
Proximity Matrix





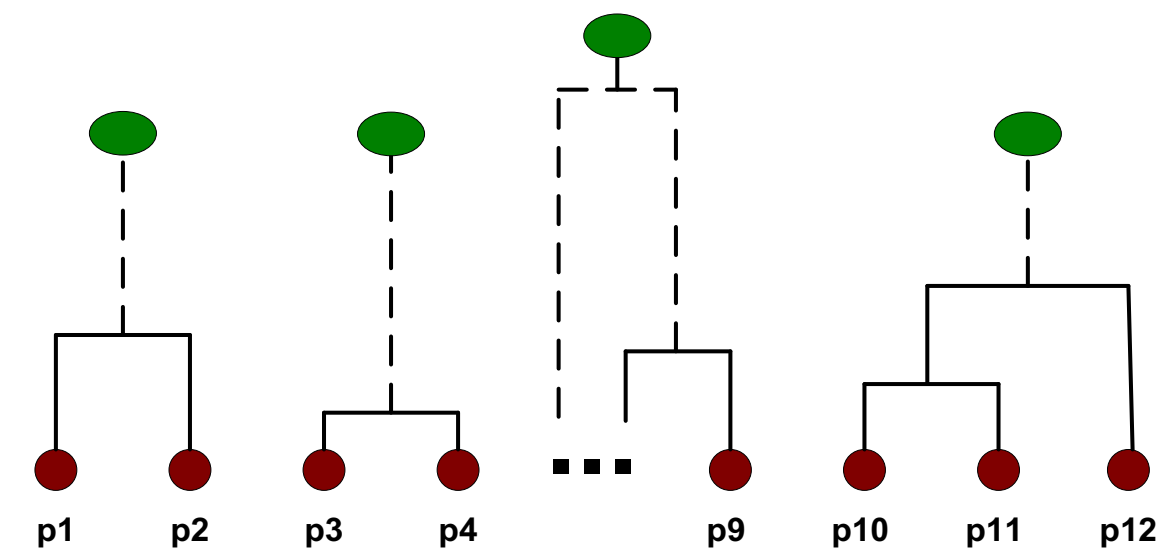
	C1	C2	C3	C4	C5
C1					
C2					
C3					
C4					
C5					

Proximity Matrix



	C1	C2 U C5	C3	C4
C1		?		
C2 U C5	?	?	?	?
C3		?		
C4		?		

Proximity Matrix



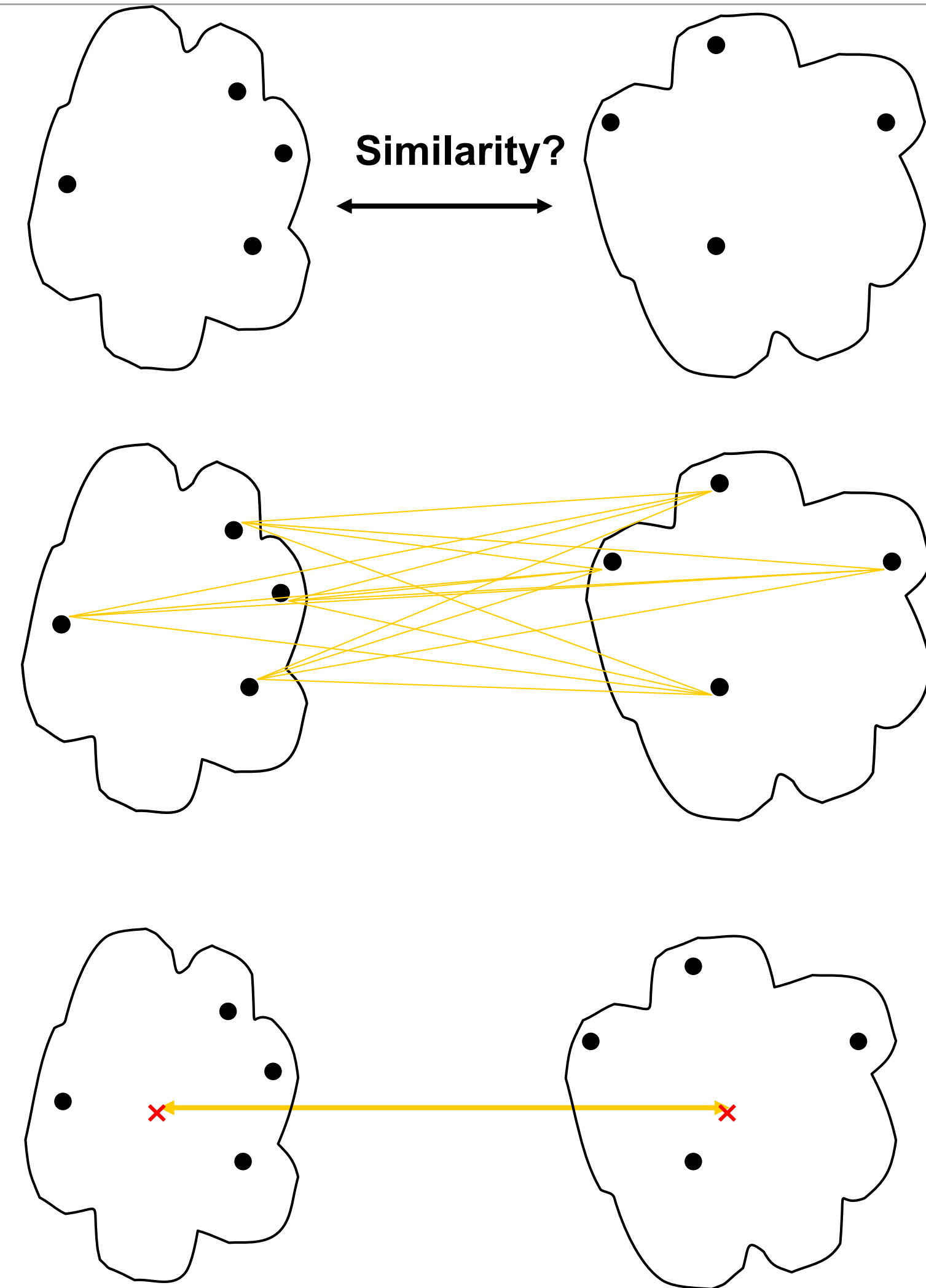
Measures of Inter-Cluster Distance

MIN (Single linkage)

MAX (Complete linkage)

Group Average (Average linkage)

Ward's Method

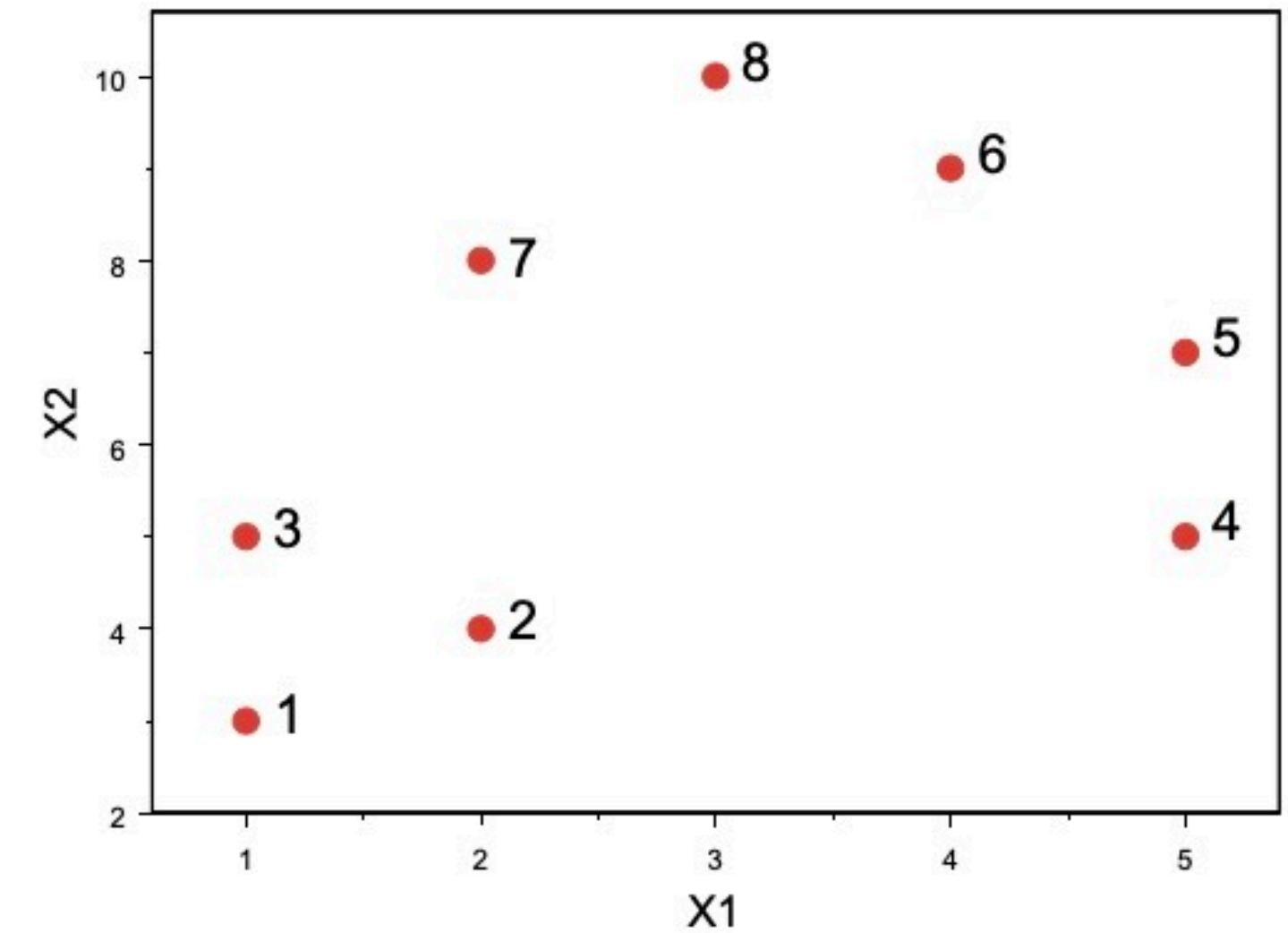


$$\mathbf{x}_1 = (1, 3)^T, \mathbf{x}_2 = (2, 4)^T, \mathbf{x}_3 = (1, 5)^T, \mathbf{x}_4 = (5, 5)^T, \\ \mathbf{x}_5 = (5, 7)^T, \mathbf{x}_6 = (4, 9)^T, \mathbf{x}_7 = (2, 8)^T, \mathbf{x}_8 = (3, 10)^T$$

Ex: Single Linkage

Pair-wise distance matrix

	1	2	3	4	5	6	7	8
1	0	1.414	2.000	4.472	5.657	6.708	5.099	7.280
2		0	1.414	3.162	4.243	5.385	4.000	6.083
3			0	4.000	4.472	5.000	3.162	5.385
4				0	2.000	4.123	4.243	5.385
5					0	2.236	3.162	3.606
6						0	2.236	1.414
7							0	2.236
8								0



Find pairs with smallest dissimilarities

$$d_{12} = d_{23} = d_{68} = 1.414$$

Merge $\mathbf{x}_2$ and $\mathbf{x}_3$ to $\{23\}$ cluster and update the proximity matrix

Updated proximity matrix

	1	23	4	5	6	7	8
1	0	1.414	4.472	5.657	6.708	5.099	7.280
23		0	3.162	4.243	5.000	3.162	5.385
4			0	2.000	4.123	4.243	5.385
5				0	2.236	3.162	3.606
6					0	2.236	1.414
7						0	2.236
8							0

Current proximity matrix

	1	23	4	5	6	7	8
1	0	1.414	4.472	5.657	6.708	5.099	7.280
23		0	3.162	4.243	5.000	3.162	5.385
4			0	2.000	4.123	4.243	5.385
5				0	2.236	3.162	3.606
6					0	2.236	1.414
7						0	2.236
8							0

Pairwise distance matrix

	1	2	3	4	5	6	7	8
1	0	1.414	2.000	4.472	5.657	6.708	5.099	7.280
2		0	1.414	3.162	4.243	5.385	4.000	6.083
3			0	4.000	4.472	5.000	3.162	5.385
4				0	2.000	4.123	4.243	5.385
5					0	2.236	3.162	3.606
6						0	2.236	1.414
7							0	2.236
8								0

Ward's Method

Merge two clusters that results in a smallest increase in the total sum of squared errors.

Let ESS_i be the SS error of every points from the centroid of cluster i .

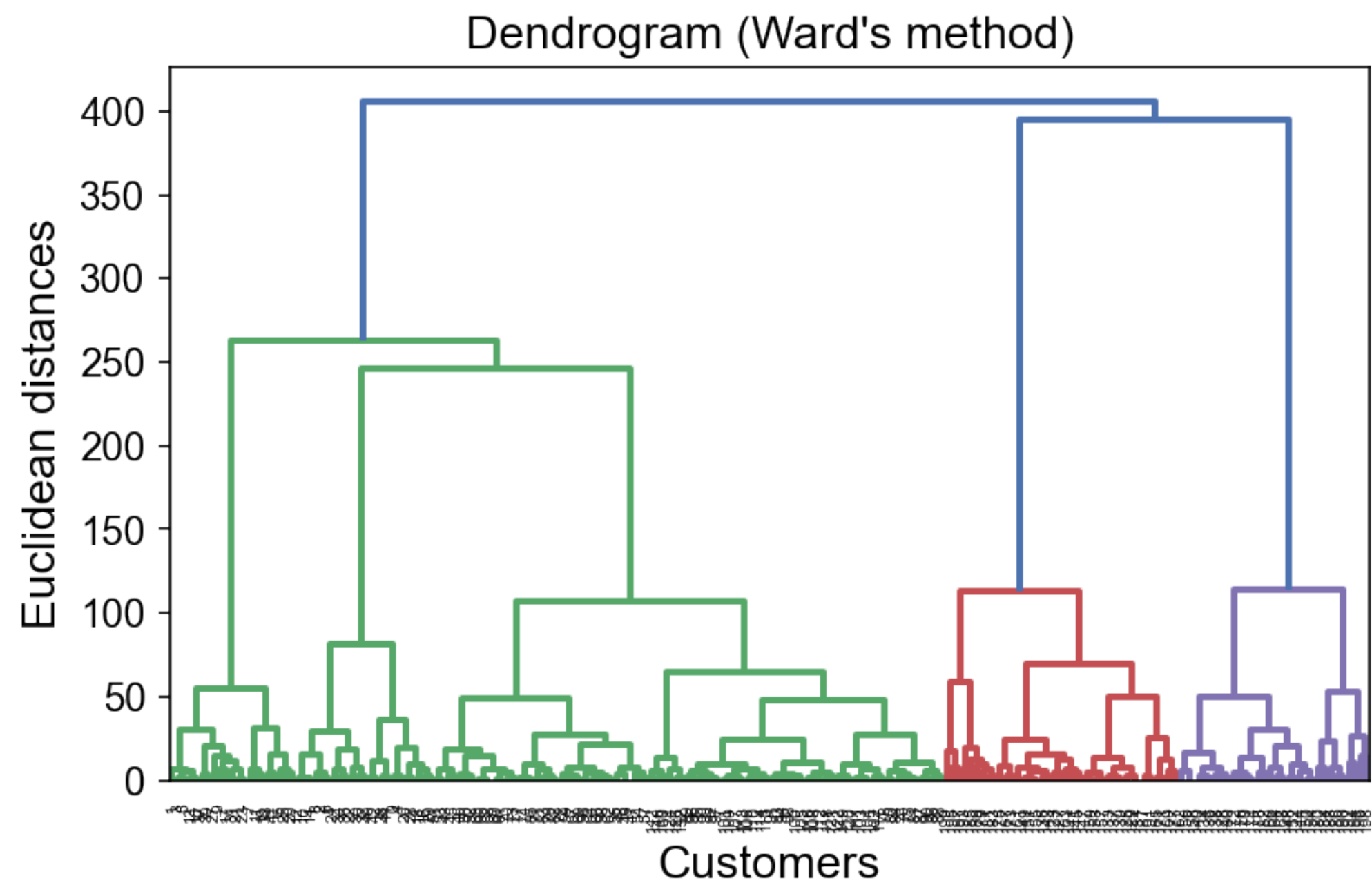
For K clusters, the total SS is $ESS_K = \sum_{i=1}^K ESS_i$

- ◆ Determine all $\binom{K}{2}$ combinations of clusters to join such that $ESS_{K-1} - ESS_K$ is smallest.
- ◆ Repeat until only one cluster is left.

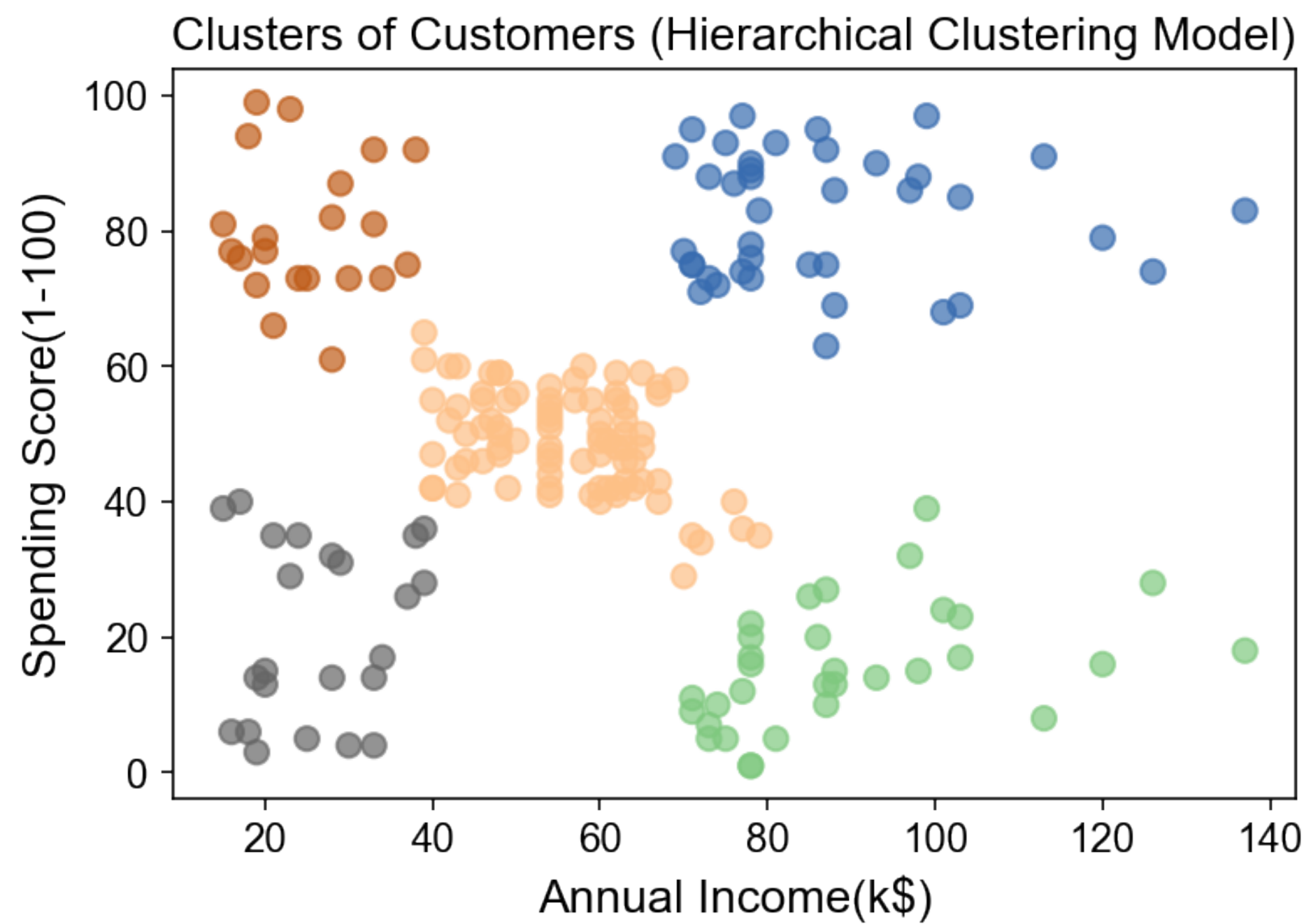
Data from sheet MALLCUST in clustering-adv.xlsx

	Gender	Age	Annual Income	SpendingScore
0	Male	19	15	39
1	Male	21	15	81
2	Female	20	16	6
3	Female	23	16	77
4	Female	31	17	40

Only three features used: Age, Annual income, Spending Score

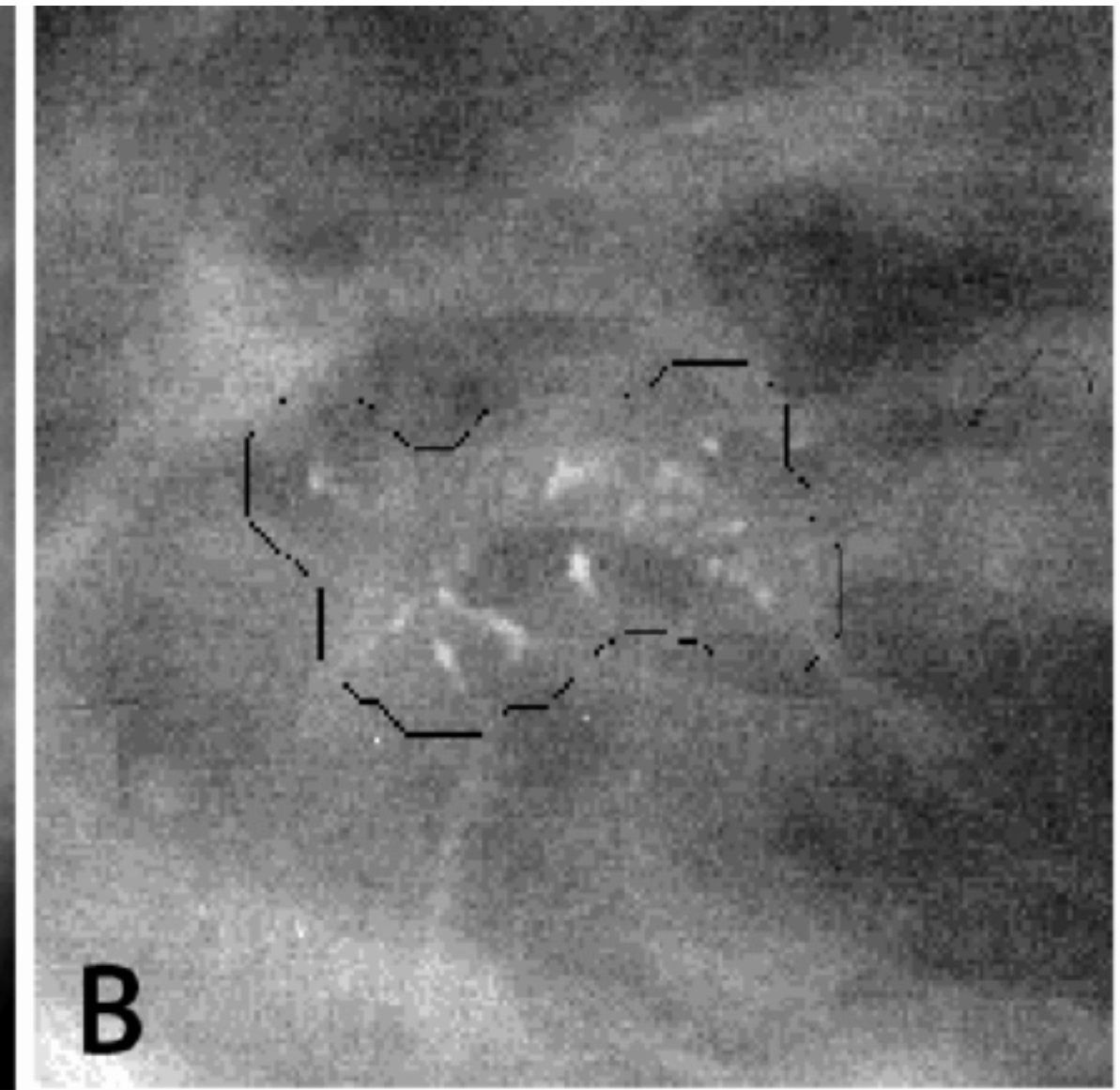
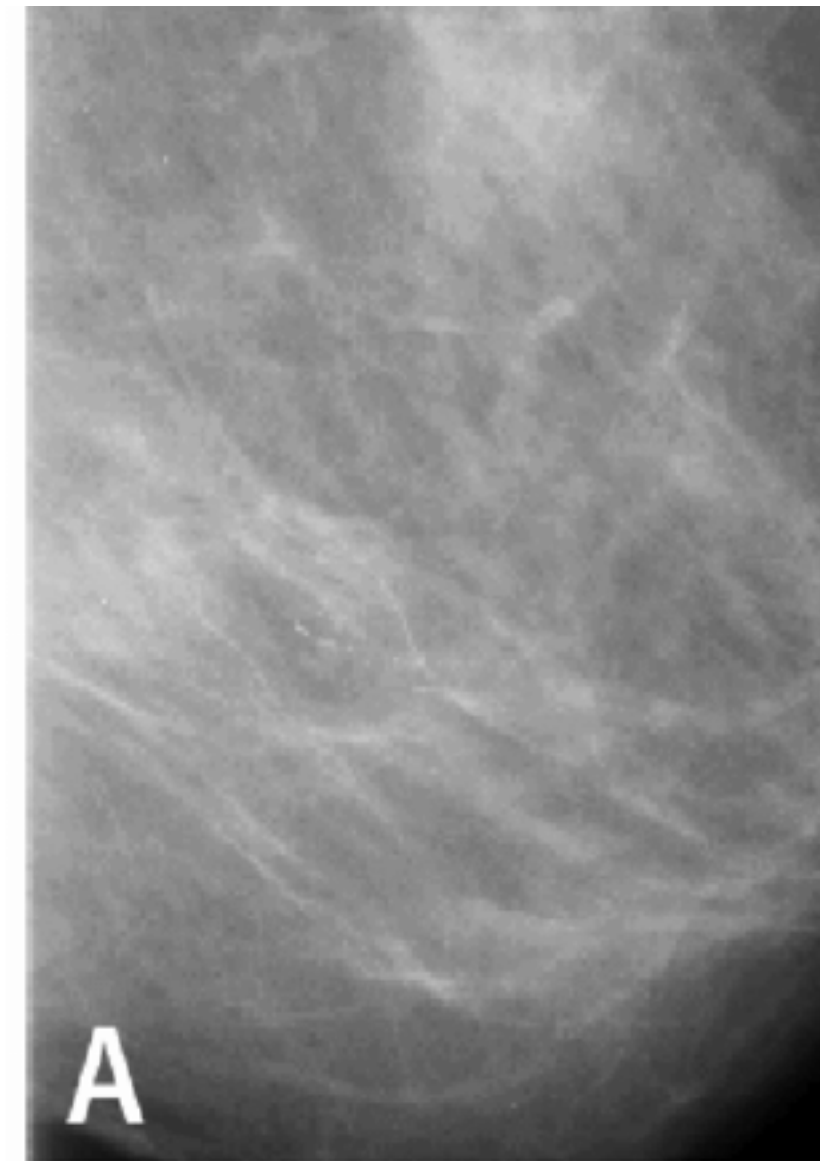
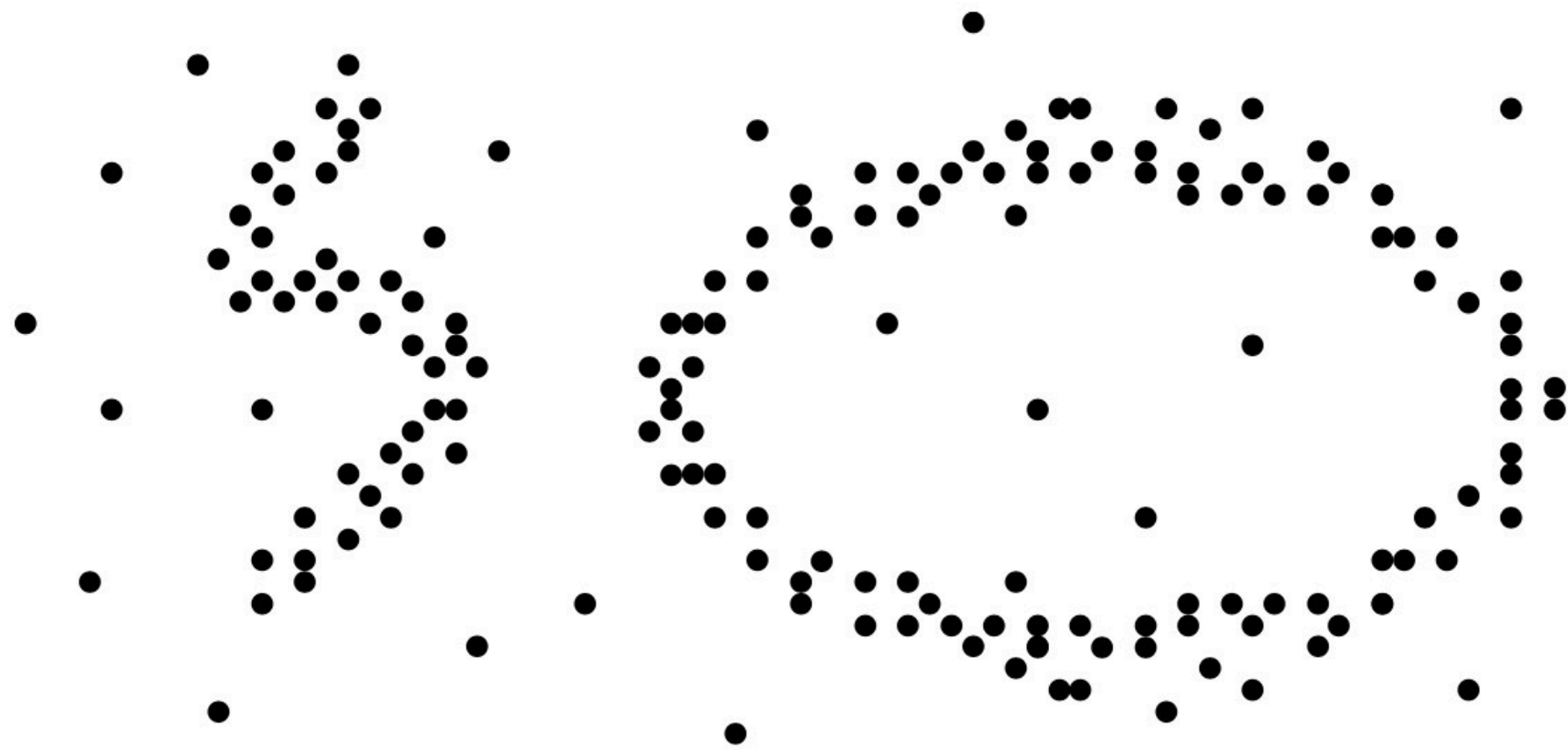


```
import scipy.cluster.hierarchy as sch
dendrogram = sch.dendrogram(sch.linkage(X, method = "ward"))
plt.title('Dendrogram (Ward\'s method)')
plt.xlabel('Customers')
plt.ylabel('Euclidean distances')
plt.tight_layout();
```



```
from sklearn.cluster import AgglomerativeClustering
hc = AgglomerativeClustering(n_clusters = 5, affinity = 'euclidean', linkage = 'ward')
y_hc=hc.fit_predict(X)
```

Density-based Clustering

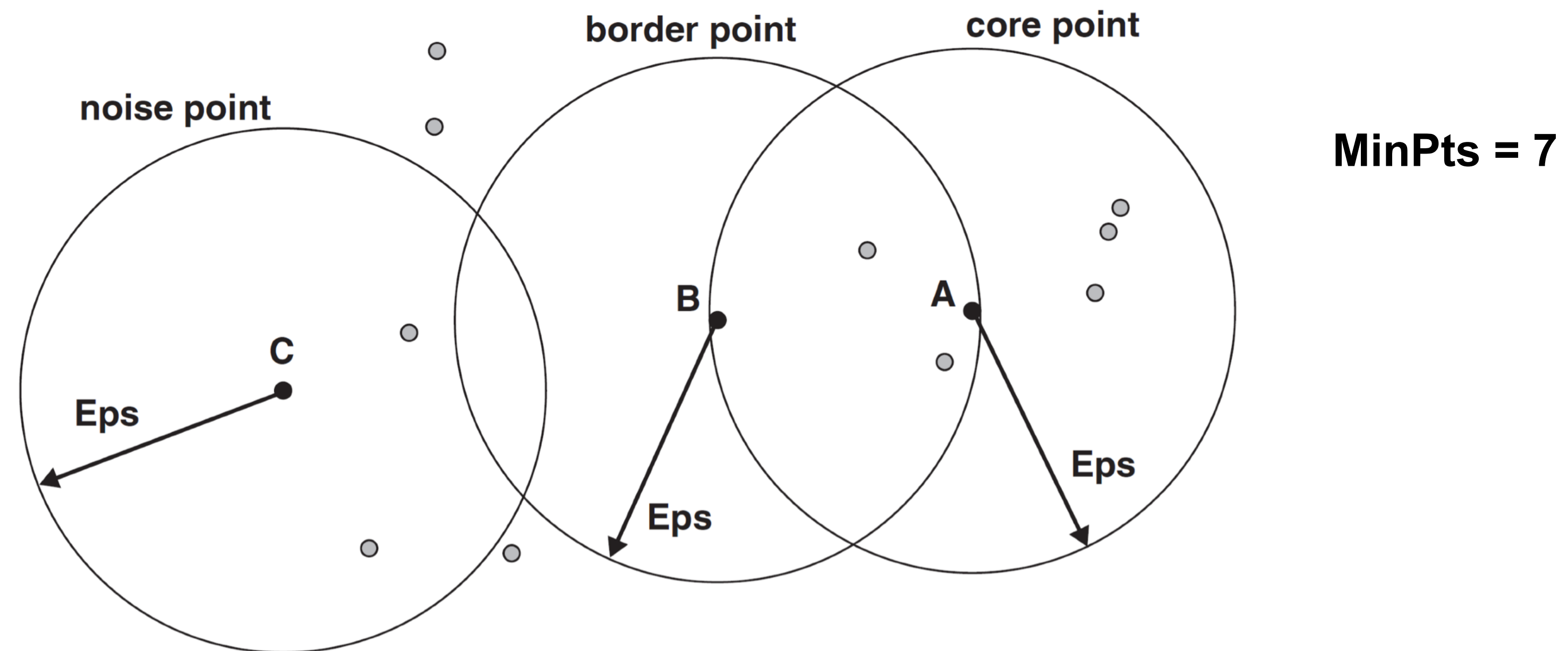


DBSCAN : Density-Based Spatial Clustering of Applications with Noise

Find objects with dense neighborhood and connect them to form dense regions as clusters.

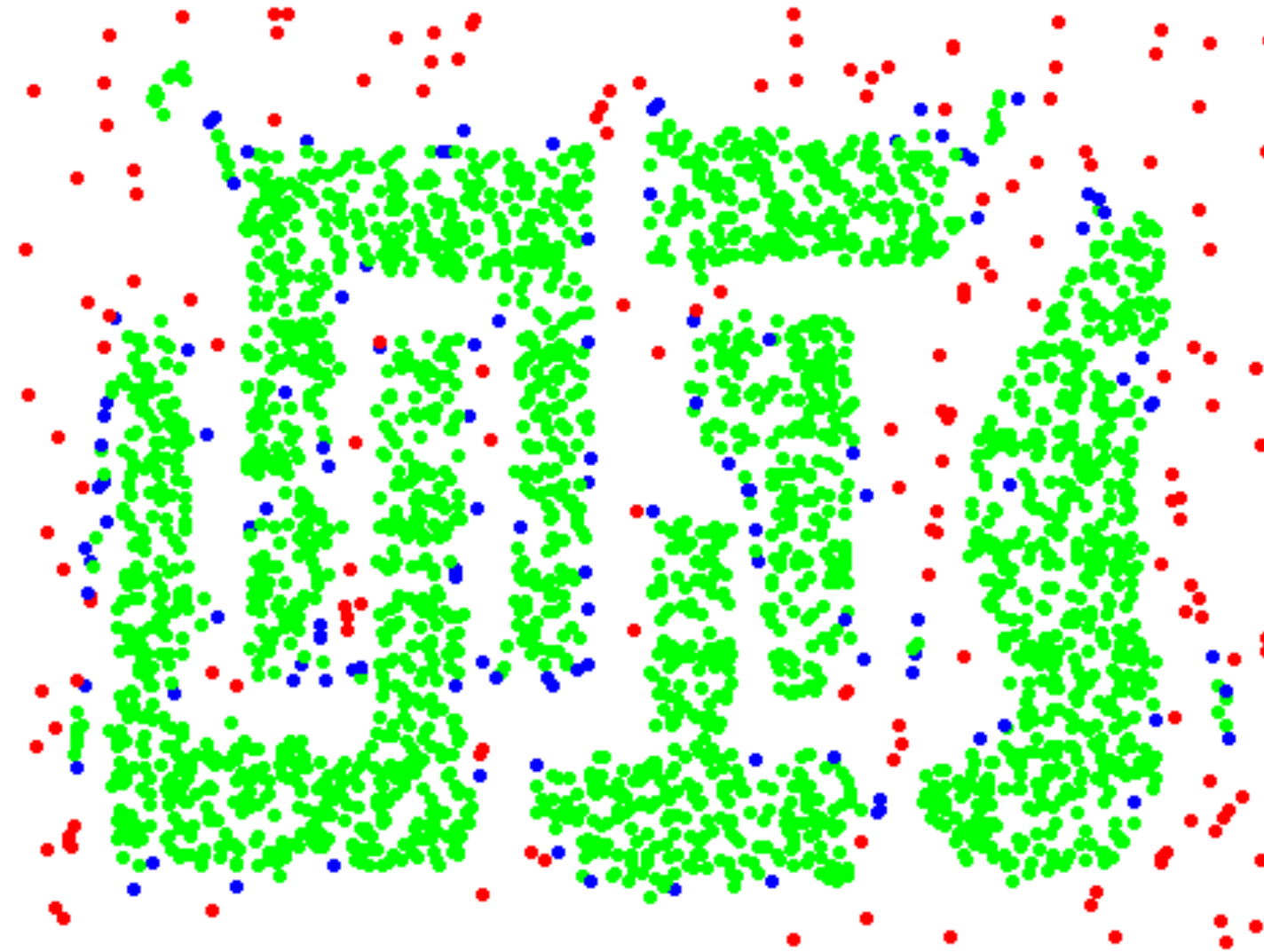
Two hyper-parameters

- ◆ Density threshold of dense region (MinPts)
- ◆ Radius of neighborhood (ϵ or Eps)



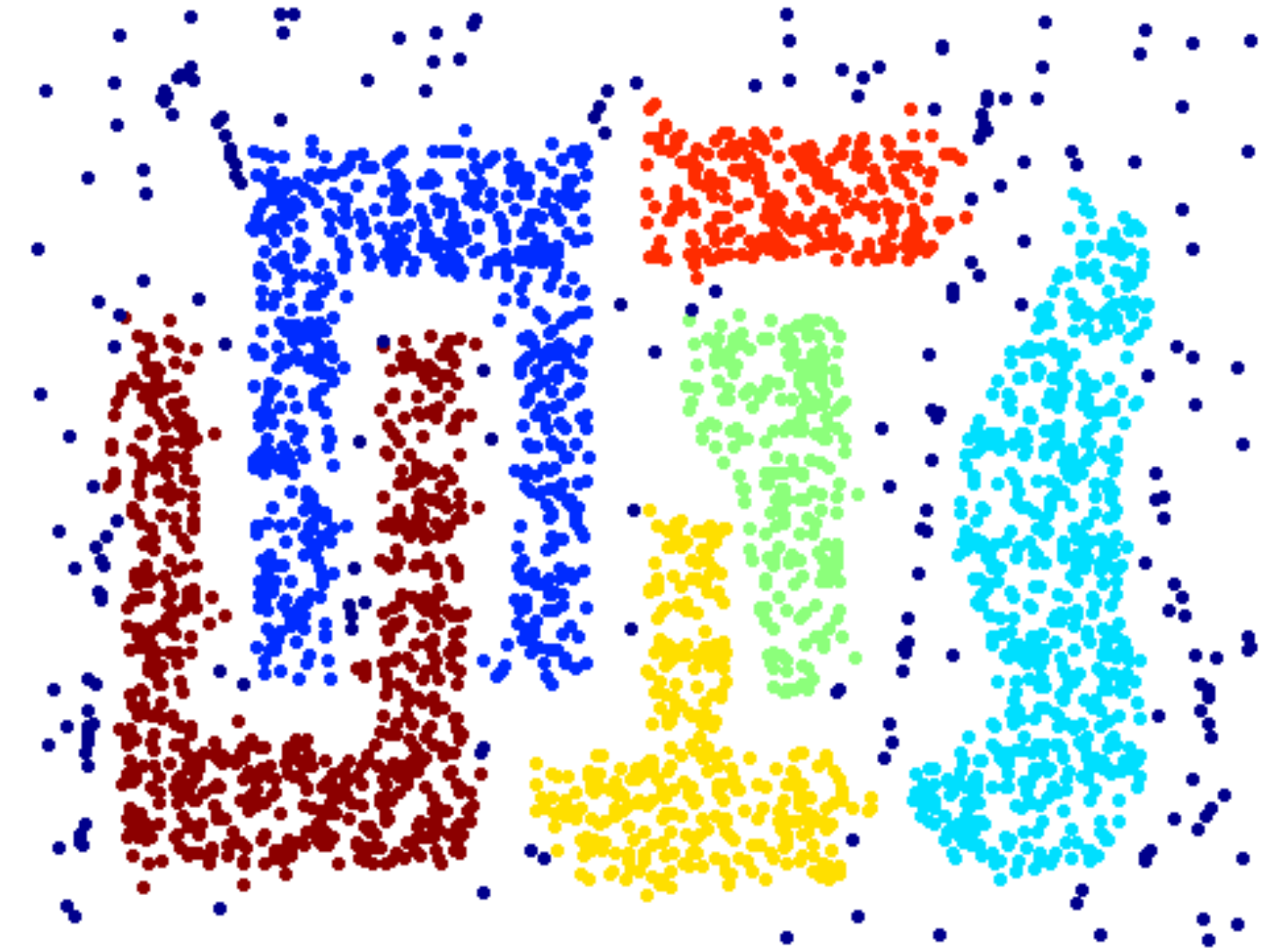


Original Points



Point types: **core**,
border and **noise**

Eps = 10, MinPts = 4



Cluster = Core points
+ Neighbors (Core/
Border points)

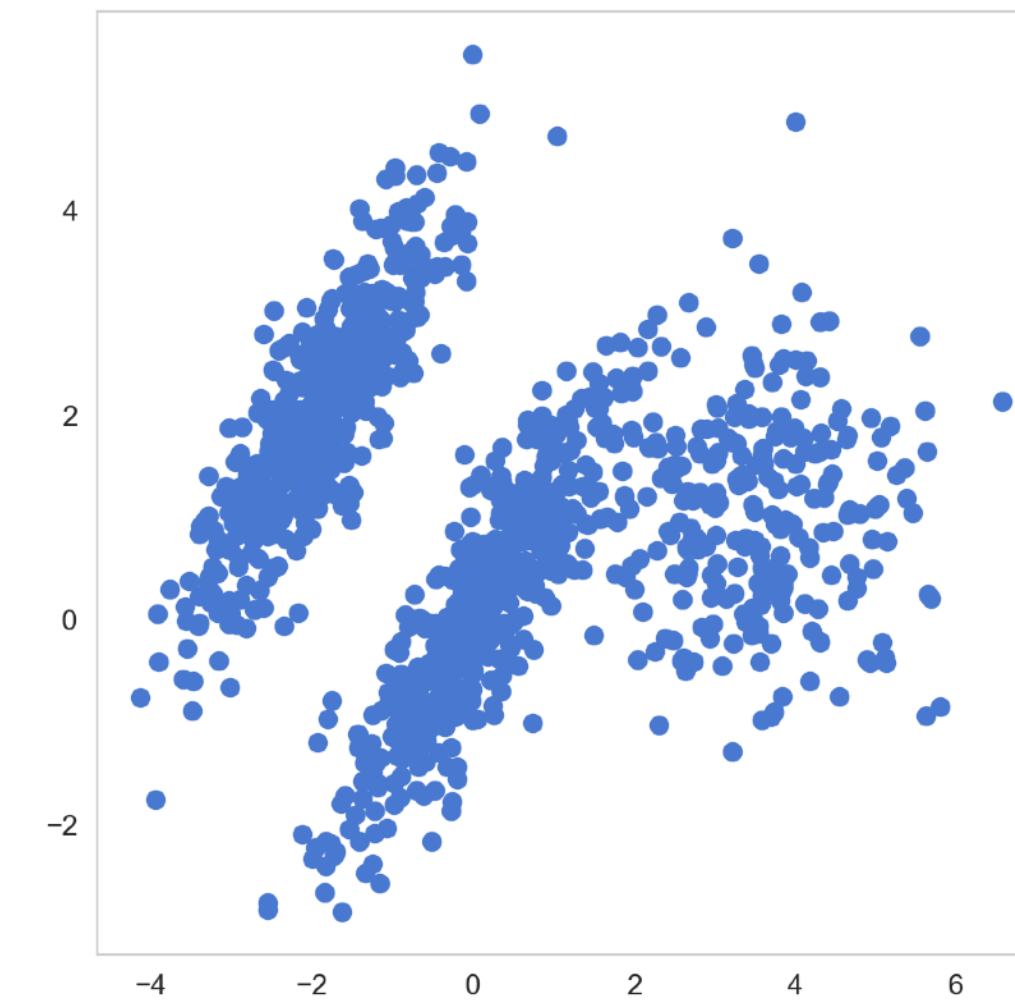
For a given MinPts, what happen if Eps is too small or too large ?

```
from sklearn.cluster import DBSCAN
```

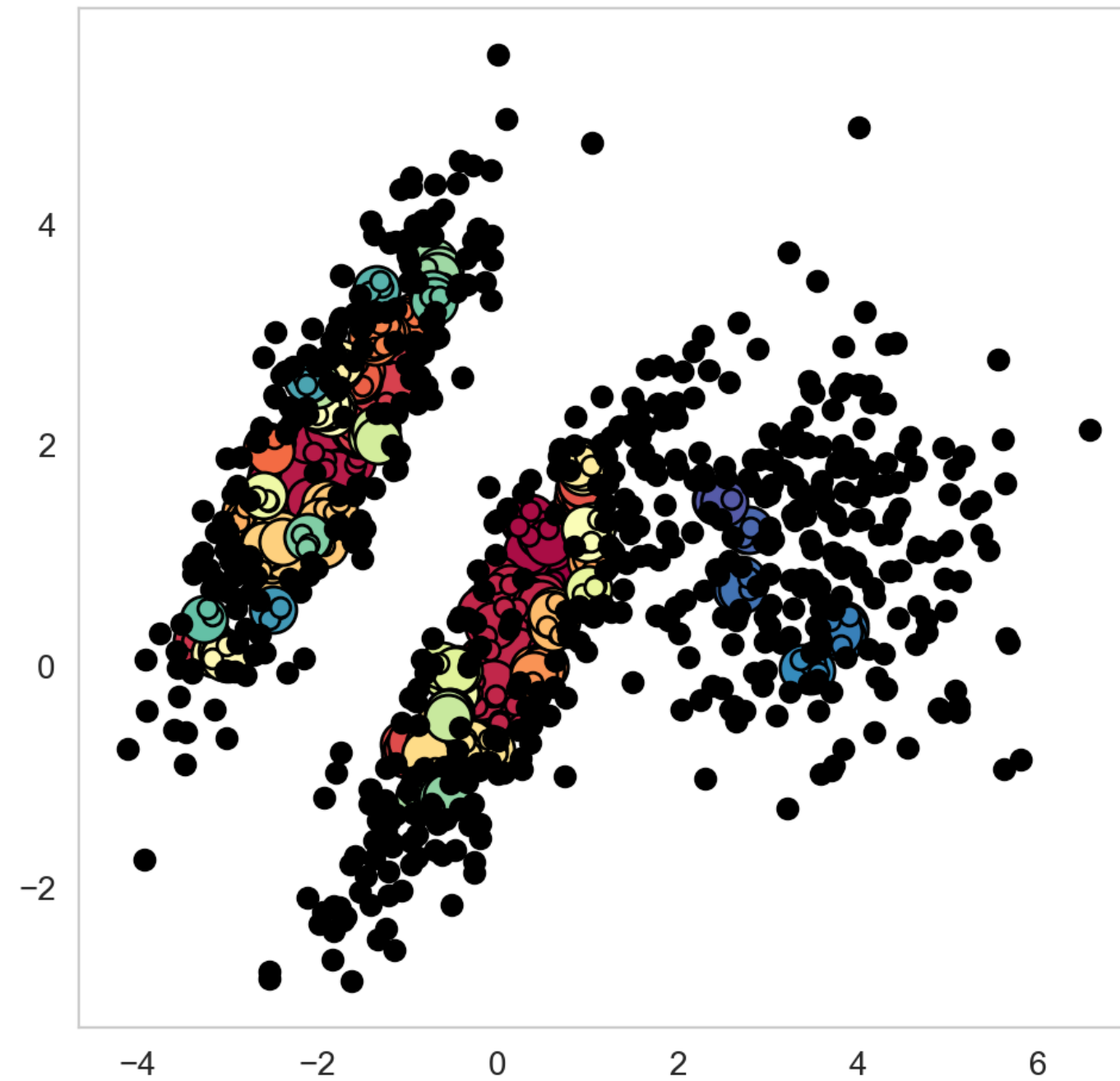
```
MinPts, Eps=4, 1.2
```

```
dbc = DBSCAN(eps=Eps, min_samples=MinPts).fit(X)
```

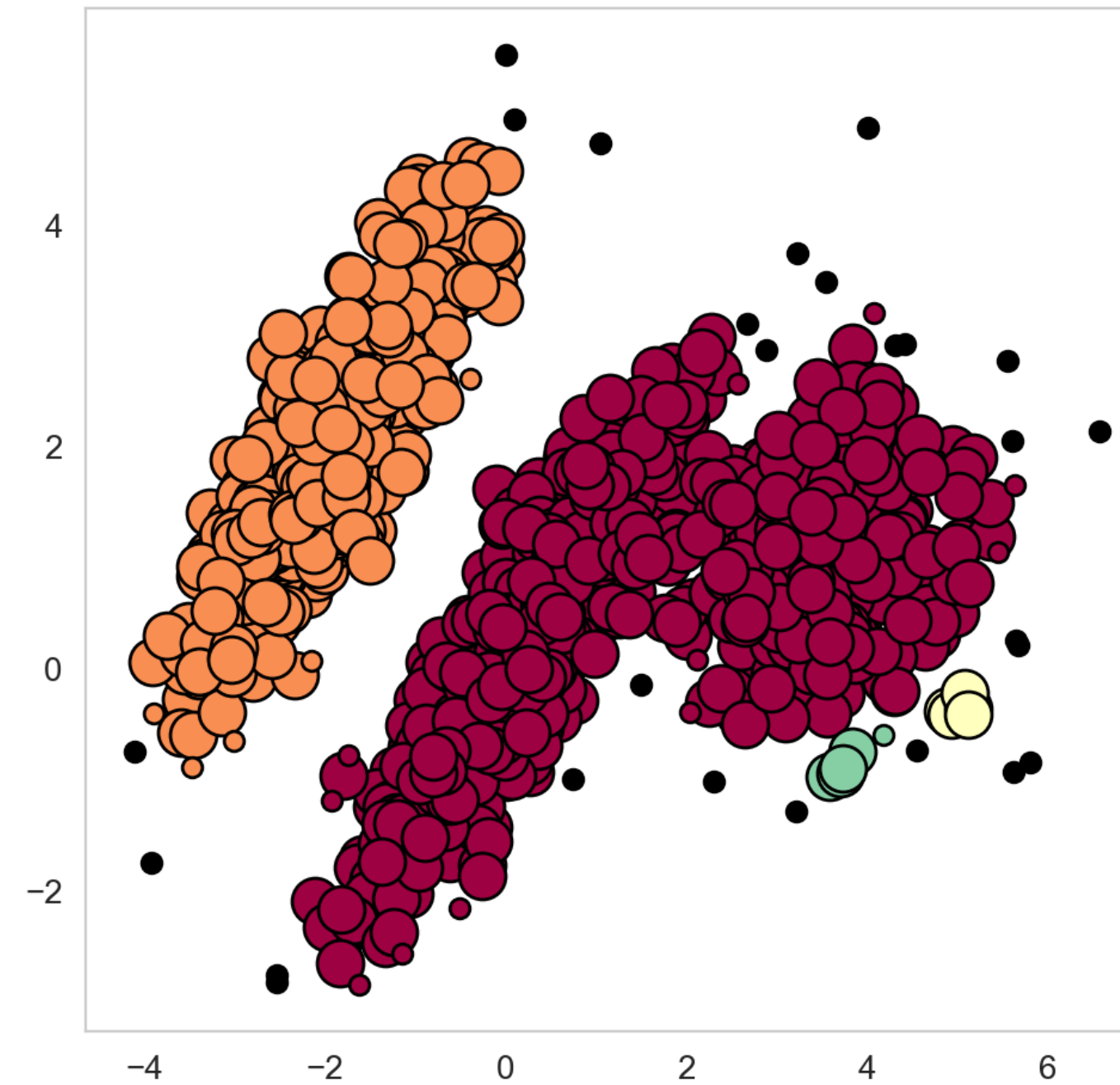
```
labels = dbc.labels_
```



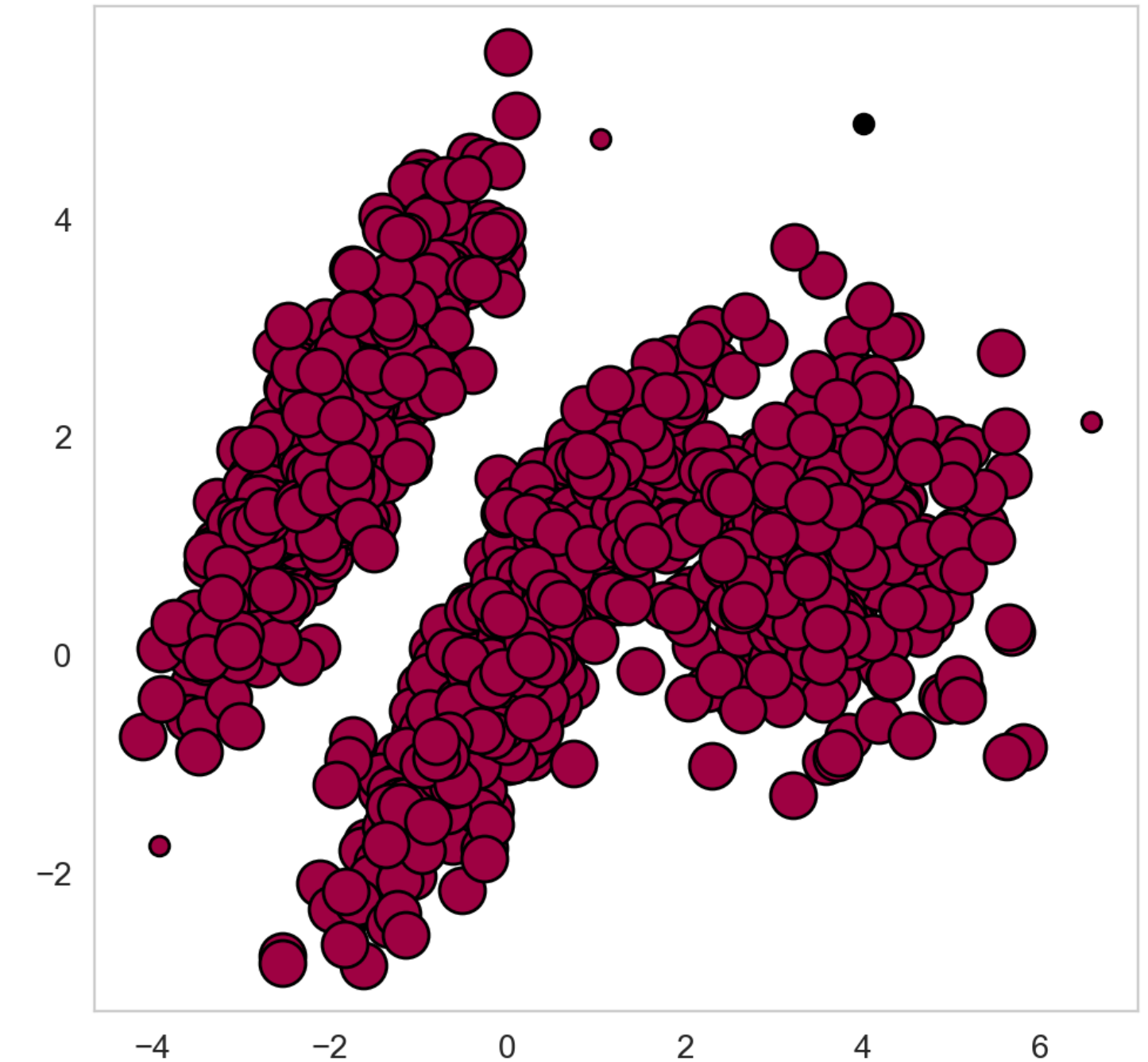
MinPts = 4, Eps=0.1, # Clusters = 46, # Noise points = 608



MinPts = 4, Eps=0.4, # Clusters = 4, # Noise points = 26



MinPts = 4, Eps=1.2, # Clusters = 1, # Noise points = 1

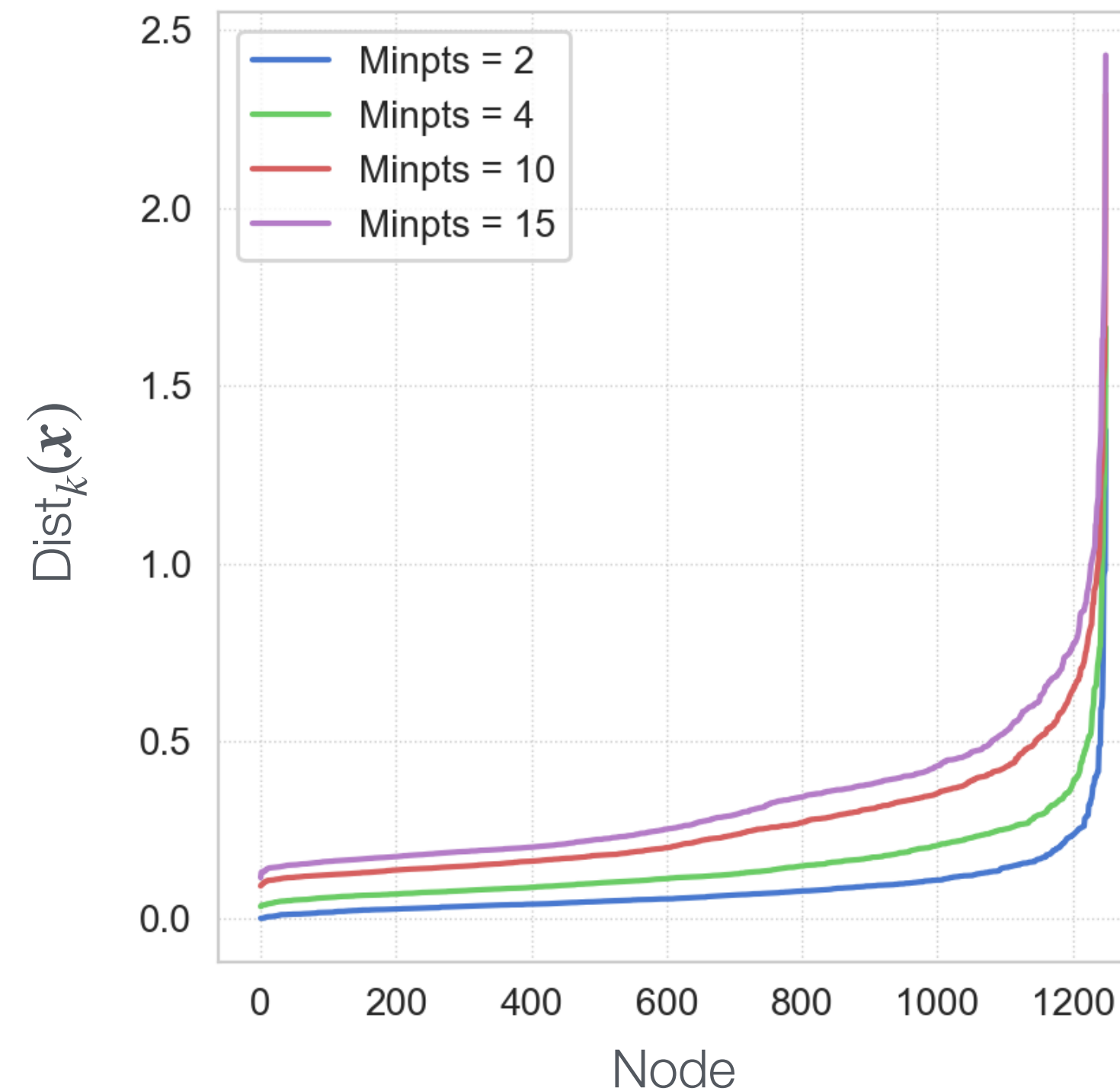
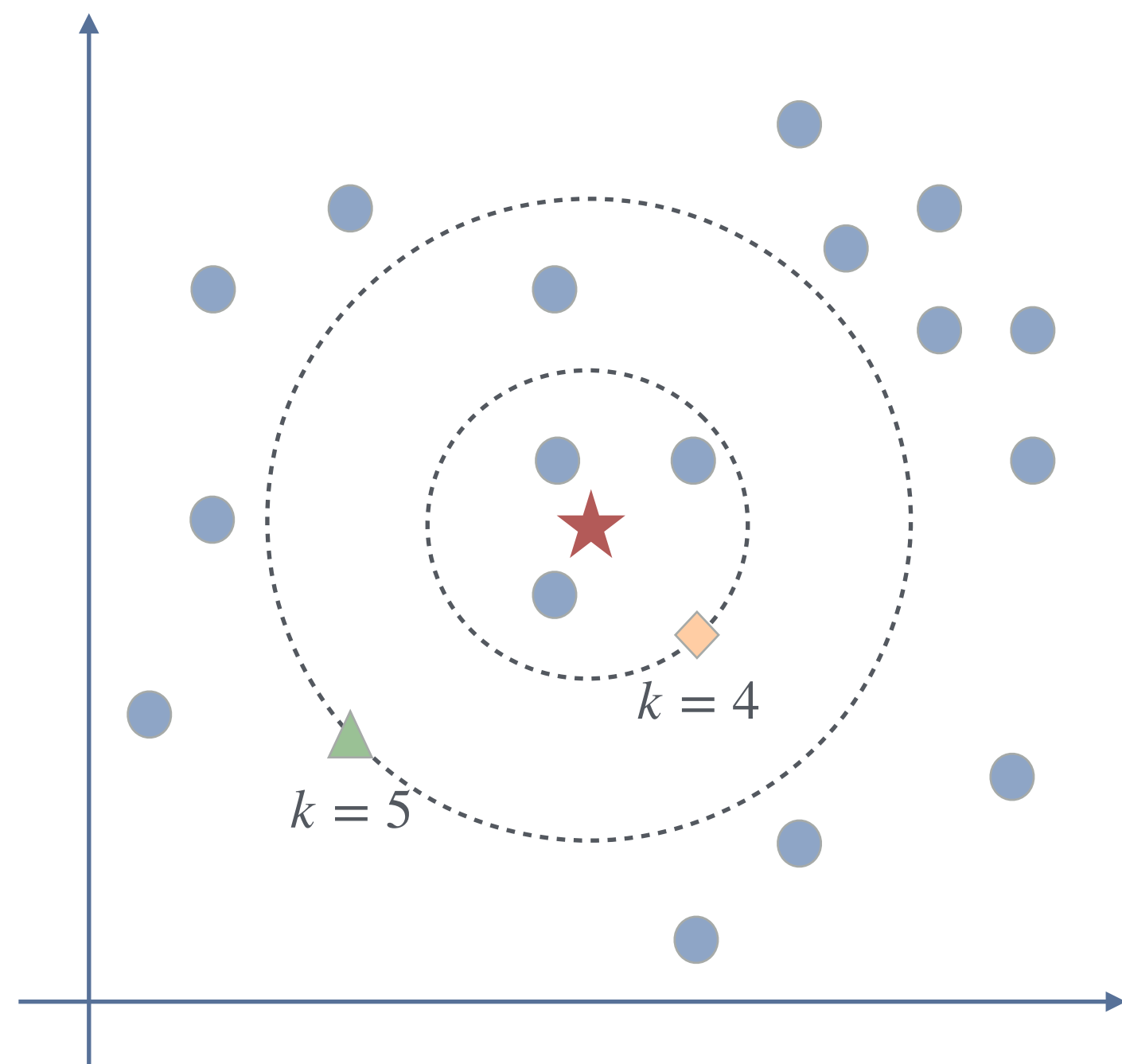


Choice of MinPts and Eps

Pick a value of MinPts and set $k = \text{MinPts}$.

For every point $\mathbf{x}$, look at $\text{dist}_k(\mathbf{x})$, distance to the k^{th} nearest neighbor.

- ◆ Small values of $\text{dist}_k(\mathbf{x})$ for points in cluster and relatively large for noise points.
- ◆ Knee point of $\text{dist}_k(\mathbf{x})$ indicates suitable value of Eps for a given $\text{MinPts} = k$.



Summary

GMM clustering

- ◆ Soft assignment based on the mixture model of Gaussian distributions
- ◆ EM algorithm to find best distribution parameters fitted to the data.
- ◆ Find elliptical-shape clusters with different sizes.

Hierarchical clustering

- ◆ Iteratively merge "closest" clusters to produces a set of nested clusters organized as a hierarchical tree.
- ◆ Clusters obtained from cutting at dendrogram at the proper level.

Density-based clustering

- ◆ Group points based on neighborhood density.
- ◆ Handle non-convex clusters and noisy data.
- ◆ Good at identifying outliers in dataset.